Data Science & AI Question Bank

Python, Statistics, ML, DL & GenAI · CrackAnalytics · by Mahendra Singh

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Python & Pandas

Q1. What is Pandas and why is it essential for data analysis? [Easy]

Pandas is Python's core data analysis library built around the **DataFrame** (table) and Series (column). It handles reading data (CSV/Excel/SQL), cleaning (nulls, types, duplicates), transformation (filtering, groupby, merge/join, pivot) and analysis — basically Excel + SQL powers inside Python, scalable to millions of rows.

```
import pandas as pd
df = pd.read_csv("sales.csv")
df.groupby("region")["amount"].sum().sort_values(ascending=False)
```

Q2. Difference between loc and iloc in Pandas? [Easy]

`loc` selects by **label** (index/column names); `iloc` selects by **integer position**.

```
df.loc[5, "sales"]      # row with index label 5, column "sales"
df.iloc[5, 2]           # 6th row, 3rd column by position
df.loc[df["region"]=="West", ["sales","profit"]]  # boolean filtering
```

Q3. How do you handle missing values in Pandas? [Medium]

```
df.isnull().sum()                 # count nulls per column
df.dropna()                       # drop rows with any null
df["age"].fillna(df["age"].median())        # fill with median
df["city"].fillna("Unknown")                # fill with constant
df["sales"] = df["sales"].interpolate()     # interpolate time series
```

Choice depends on meaning: drop when few and random; median/mode-fill for skewed numeric/categorical; interpolate for time series; sometimes a "missing" flag column is itself informative.

Q4. Difference between merge, join and concat in Pandas? [Medium]

`pd.merge()` — SQL-style join on key columns (how = inner/left/right/outer). `df.join()` — convenience join on the index. `pd.concat()` — stacks DataFrames vertically (rows, like UNION ALL) or horizontally.

```
pd.merge(orders, customers, on="customer_id", how="left")
pd.concat([jan_df, feb_df, mar_df])
```

Q5. Explain groupby with an example — the analyst's daily tool [Medium]

```
df.groupby("region").agg(
    total_sales=("amount", "sum"),
    avg_order  =("amount", "mean"),
    orders     =("order_id", "nunique")
).reset_index()
```

Split-apply-combine: split rows into groups, apply aggregations, combine into a summary table — the Pandas equivalent of GROUP BY in SQL / a pivot table in Excel.

Q6. How do you remove duplicates and find them first? [Easy]

```
df.duplicated().sum()                         # how many duplicate rows
df[df.duplicated(subset=["email"], keep=False)]  # view all dup emails
df = df.drop_duplicates(subset=["email"], keep="first")
```

Q7. What is a lambda function and where do analysts use it? [Easy]

An anonymous one-line function, mostly with `apply()` for quick row/column transformations:

```
df["price_band"] = df["price"].apply(
    lambda x: "High" if x > 1000 else "Low")
```

For pure element-wise math, vectorized operations (`df["a"]*df["b"]`) are faster than apply+lambda.

Q8. Difference between a list, tuple, set and dictionary? [Easy]

- **List** [1,2,2] — ordered, mutable, allows duplicates.
- **Tuple** (1,2) — ordered, **immutable**; used for fixed records, dict keys.
- **Set** {1,2} — unordered, unique values; fast membership tests, dedup.
- **Dict** {"a":1} — key→value mapping; the workhorse for lookups and JSON-like data.

Q9. How would you read a 10GB CSV that doesn't fit in memory? *[Hard]*

```
chunks = pd.read_csv("big.csv", chunksize=1_000_000)
total = sum(chunk["amount"].sum() for chunk in chunks)
```

Options: **chunked reading** and aggregating per chunk; loading only needed columns (`usecols`) with efficient dtypes; converting to Parquet; or scaling out with DuckDB/Polars/Dask/Spark for genuinely big data.

Statistics & Probability

Q1. Mean vs median vs mode — when does median beat mean? *[Easy]*

Mean = average; median = middle value; mode = most frequent. **Median wins with skewed data/outliers**: for salaries [30k, 35k, 40k, 45k, 10L], mean ≈ 2.3L (misleading) while median = 40k (representative). That's why house prices and incomes are reported as medians.

Q2. Explain standard deviation and variance simply *[Easy]*

Both measure spread around the mean. Variance = average of squared deviations; standard deviation = its square root (same units as the data). Two shops can both average ■50k daily sales — SD 2k means steady, SD 30k means wild swings. Low SD = consistent, high SD = volatile.

Q3. Correlation vs causation — the classic trap *[Medium]*

Correlation measures how two variables move together (–1 to +1); causation means one *drives* the other. Ice-cream sales correlate with drownings — summer causes both. Analysts must say "associated with", check confounders, and only claim causation from controlled experiments (A/B tests).

Q4. What is a p-value in plain language? *[Medium]*

The probability of seeing results at least this extreme **if there were truly no effect** (null hypothesis true). p = 0.03 → only a 3% chance this pattern is pure luck → at the common 0.05 threshold we call it statistically significant. It is NOT the probability the hypothesis is true, and significance ≠ business importance.

Q5. Explain hypothesis testing with a business example *[Medium]*

Question: did the new checkout page increase conversion? **H0** (null): no difference. **H1**: conversion increased. Run an A/B test, compute the test statistic (e.g. two-proportion z-test), get the p-value; if $p < 0.05$ reject H0 and roll out the new page. Also check **sample size/power** before trusting the result.

Q6. What is the Central Limit Theorem and why does it matter? *[Hard]*

CLT: the distribution of **sample means** approaches a normal curve as sample size grows (~30+), regardless of the population's shape. It's why we can build confidence intervals and run t-tests on revenue-per-user or delivery times even when the raw data is skewed.

Q7. Type I vs Type II error? *[Hard]*

Type I (false positive): rejecting a true null — claiming the campaign worked when it didn't (probability = α, usually 5%). **Type II (false negative):** missing a real effect (probability = β; power = $1-\beta$). Business framing: Type I wastes money on a fake win; Type II leaves a real win on the table.

Q8. What are outliers and how do you detect & treat them? *[Medium]*

Values far from the rest. Detect: **IQR rule** (outside Q1–1.5·IQR to Q3+1.5·IQR), z-score > 3, or box plots. Treat: investigate first (data-entry error vs genuine event) → fix errors, cap/winsorize, analyze with and without, or use robust metrics (median). Never silently delete — a fraud spike "outlier" may be the whole story.

Machine Learning

Q1. What is Machine Learning in one interview-ready line? *[Easy]*

ML is teaching computers to learn patterns from data and make predictions/decisions **without being explicitly programmed with rules** — e.g. learning from past transactions which future ones look fraudulent.

Q2. Supervised vs Unsupervised vs Reinforcement learning? [Easy]

- **Supervised:** learn from labelled data (X → known Y). Predict price, detect spam. Algorithms: linear/logistic regression, decision trees, random forest, XGBoost.
- **Unsupervised:** find structure in unlabelled data. Customer segmentation (K-Means), anomaly detection, PCA.
- **Reinforcement:** an agent learns by trial-and-error rewards — game AI, robotics, recommendation tuning.

Q3. Regression vs Classification? [Easy]

Both supervised. **Regression** predicts a continuous number (next month's sales, house price). **Classification** predicts a category (churn: yes/no, sentiment: positive/neutral/negative). Same data can frame both: predict revenue (regression) vs predict "will spend >■10k?" (classification).

Q4. Explain overfitting and how to prevent it [Medium]

Overfitting = the model memorizes training data (noise included) and fails on new data — 99% train accuracy, 65% test. Prevent: **train/test split & cross-validation**, simpler models, regularization (L1/L2), pruning/limiting tree depth, more data, early stopping, dropout (neural nets). Underfitting is the opposite — too simple to capture the pattern.

Q5. What is the bias–variance tradeoff? [Hard]

Bias = error from oversimplifying (underfit); **variance** = error from oversensitivity to training data (overfit). Simple models: high bias/low variance; complex models: low bias/high variance. The art is the sweet spot in the middle — tuned via validation curves and regularization.

Q6. Explain train-test split and cross-validation [Medium]

Split data (e.g. 80/20) so the model is **evaluated on unseen data**. K-fold cross-validation goes further: split into K parts, train K times each holding out one fold, average the scores — a more reliable estimate, especially on small datasets. Golden rule: test data must never leak into training (fit scalers/encoders on train only).

Q7. Precision vs Recall vs F1 — and when accuracy lies [Hard]

With 99% legit transactions, a model saying "never fraud" is 99% accurate and useless. **Precision** = of predicted positives, how many were right (cost of false alarms). **Recall** = of actual positives, how many we caught (cost of missing). **F1** = harmonic mean of both. Fraud/cancer screening → prioritize recall; spam filtering → precision.

Q8. Explain Linear and Logistic Regression simply [Medium]

Linear regression fits a line y = b0 + b1x to predict a number (ad spend → sales). **Logistic regression** passes that line through a sigmoid to output a 0–1 probability for classification (will the customer churn?). Despite the name, logistic regression is a classification algorithm — and a favourite interview trick question.

Q9. How does a Decision Tree work, and what is a Random Forest? [Medium]

A decision tree splits data by the most informative questions ("Income > 50k?") into purer and purer groups — easy to explain, prone to overfitting. A **random forest** builds hundreds of trees on random data/feature subsets and votes — much more accurate and stable. Follow-up: **bagging** (forest, parallel) vs **boosting** (XGBoost, sequential error-fixing).

Q10. What is K-Means clustering? Give a business use case [Medium]

Unsupervised algorithm grouping data into K clusters: pick K centers → assign points to nearest center → recompute centers → repeat until stable. Use case: **customer segmentation** on recency/frequency/monetary value revealing "champions", "at-risk", "bargain hunters" for targeted marketing. K is chosen via the elbow method/silhouette score.

Q11. What is feature engineering? Examples [Medium]

Creating better model inputs from raw data — often more impactful than changing algorithms. Examples: date → day-of-week/month/festival-flag; amount → log(amount) for skew; address → distance-from-city-center; transactions → RFM features per customer; categorical → one-hot/target encoding; scaling for distance-based models.

Generative AI & LLMs

Q1. What is Generative AI and how is it different from traditional ML? [Easy]

Traditional ML **predicts** (a number, a class) from patterns; Generative AI **creates new content** — text, images, code — by learning the underlying distribution of data. LLMs like GPT and Claude generate text token by token, predicting the next most likely token given context, trained on massive corpora.

Q2. What is an LLM and what does 'token' mean? [Easy]

A Large Language Model is a neural network (transformer architecture) with billions of parameters trained on huge text datasets to predict the next token. A **token** is a chunk of text (~4 characters / ¾ of a word in English) — models read and generate token by token, and context windows and API pricing are measured in tokens.

Q3. What is prompt engineering? Give practical techniques [Medium]

Designing inputs to get reliably good LLM outputs. Techniques: be specific with role + task + format ("You are a SQL expert; return only the query"); give **few-shot examples**; ask for step-by-step reasoning on complex problems; constrain output (JSON schema, word limits); iterate. As an analyst: "Write a SQL query for monthly sales by region from table X with columns A, B, C" beats "help with SQL".

Q4. What are hallucinations in LLMs and how do you mitigate them? [Medium]

Confident but false outputs — invented statistics, fake citations, wrong formulas. Mitigate: ground the model with **RAG** (retrieval of real documents), ask for sources, lower temperature for factual tasks, validate critical outputs (run the SQL, check the number), and human review. Analysts should treat LLM output as a smart draft, not truth.

Q5. What is RAG (Retrieval-Augmented Generation)? [Hard]

Architecture that fixes an LLM's knowledge limits: user query → **retrieve** relevant chunks from your documents (via embeddings + vector database) → inject them into the prompt → LLM answers **grounded in your data**. This is how "chat with your company docs/PDF" products work — no retraining needed, answers stay current and citable.

Q6. What are embeddings? Why do analysts care? [Hard]

Numeric vector representations of text/items where similar meanings sit close together — "refund not received" ≈ "money not returned". Uses: semantic search, clustering customer feedback into themes, deduplication, recommendation, powering RAG. They let you do math on meaning.

Q7. How is AI changing the data analyst role? (common HR + tech question) [Easy]

Balanced answer: AI accelerates the mechanical parts — writing SQL/DAX drafts, cleaning scripts, chart suggestions, summarizing findings — so analysts shift up the value chain: framing the right business questions, validating AI output, data quality, storytelling and decisions. "I use AI as a productivity multiplier, but I verify everything because I own the accuracy."

Q8. What is the difference between AI, ML and Deep Learning? [Easy]

Nested circles: **AI** = any technique making machines act intelligently → **ML** = the subset that learns from data instead of hard-coded rules → **Deep Learning** = the subset of ML using multi-layer neural networks (images, speech, LLMs). Every LLM is DL, every DL is ML, every ML is AI — not vice versa.

ML & Data Science practice bank

Q1. What's the trade-off between bias and variance?

Bias is error due to erroneous or overly simplistic assumptions in the learning algorithm you're using. This can lead to the model underfitting your data, making it hard for it to have high predictive accuracy and for you to generalize your knowledge from the training set to the test set.

Variance is error due to too much complexity in the learning algorithm you're using. This leads to the algorithm being highly sensitive to high degrees of variation in your training data, which can lead your model to overfit the data. You'll be carrying too much noise from your training data for your model to be very useful for your test data.

The bias-variance decomposition essentially decomposes the learning error from any algorithm by adding the bias, the variance and a bit of irreducible error due to noise in the underlying dataset. Essentially, if you make the model more complex and add more variables, you'll lose bias but gain some variance — in order to get the optimally reduced amount of error, you'll have to tradeoff bias and variance. You don't want either high bias or high variance in your model.

Q2. What is the difference between supervised and unsu- pervised machine learning?

vised machine learning? (Quora)

Supervised learning requires training labeled data. For example, in order to do classification (a supervised learning task), you'll need to first label the data you'll use to train the model to classify data into your labeled groups. Unsupervised learning, in contrast, does not require labeling data explicitly.

Q3. How is KNN different from k-means clustering?

k-means clustering? (Quora)

K-Nearest Neighbors is a supervised classification algorithm, while k-means clustering is an unsupervised clustering algorithm. While the mechanisms may seem similar at first, what this really means is that in order for K-Nearest Neighbors to work, you need labeled data you want to classify an unlabeled point into (thus the nearest neighbor part). K-means clustering requires only a set of unlabeled points and a threshold: the algorithm will take unlabeled points and gradually learn how to cluster them into groups by computing the mean of the distance between different points.

The critical difference here is that KNN needs labeled points and is thus supervised learning, while k-means doesn't — and is thus unsupervised learning.

Q4. What is Bayes' Theorem? How is it useful in a machine learning context?

(BetterExplained)

Bayes' Theorem gives you the posterior probability of an event given what is known as prior knowledge.

Mathematically, it's expressed as the true positive rate of a condition sample divided by the sum of the false positive rate of the population and the true positive rate of a condition. Say you had a 60% chance of actually having the flu after a flu test, but out of people who had the flu, the test will be false 50% of the time, and the overall population only has a 5% chance of having the flu. Would you actually have a 60% chance of having the flu after having a positive test?

Bayes' Theorem says no. It says that you have a (.6 * 0.05) (True Positive Rate of a Condition Sample) / (.6*0.05)(True Positive Rate of a Condition Sample) + (.5*0.95) (False Positive Rate of a Population) = 0.0594 or 5.94% chance of getting a flu.

Bayes' Theorem is the basis behind a branch of machine learning that most notably includes the Naive Bayes classifier. That's something important to consider when you're faced with machine learning interview questions.

Q5. Why is "Naive" Bayes naive?

Despite its practical applications, especially in text mining, Naive Bayes is considered "Naive" because it makes an assumption that is virtually impossible to see in real-life data: the conditional probability is calculated as the pure product of the individual probabilities of components. This implies the absolute independence of features — a condition probably never met in real life.

As a Quora commenter put it whimsically, a Naive Bayes classifier that figured out that you liked pickles and ice cream would probably naively recommend you a pickle ice cream.

Q6. What's your favorite algorithm, and can you explain it to me in less than a minute?

This type of question tests your understanding of how to communicate complex and technical nuances with poise and the ability to summarize quickly and efficiently. Make sure you have a choice and make sure you can explain different algorithms so simply and effectively that a five-year-old could grasp the basics!

Q7. What's the difference between Type I and Type II error?

Don't think that this is a trick question! Many machine learning interview questions will be an attempt to lob basic questions at you just to make sure you're on top of your game and you've prepared all of your bases.

Type I error is a false positive, while Type II error is a false negative. Briefly stated, Type I error means claiming something has happened when it hasn't, while Type II error means that you claim nothing is happening when in fact something is.

A clever way to think about this is to think of Type I error as telling a man he is pregnant, while Type II error means you tell a pregnant woman she isn't carrying a baby.

Q8. What's a Fourier transform?

A Fourier transform is a generic method to decompose generic functions into a superposition of symmetric functions. Or as this more intuitive tutorial puts it, given a smoothie, it's how we find the recipe. The Fourier transform finds the set of cycle speeds, amplitudes and phases to match any time signal. A Fourier transform converts a signal from time to frequency domain — it's a very common way to extract features from audio signals or other time series such as sensor data.

Q9. What is deep learning, and how does it contrast with other machine learning algorithms?

Deep learning is a subset of machine learning that is concerned with neural networks: how to use backpropagation and certain principles from neuroscience to more accurately model large sets of unlabelled or semi-structured data. In that sense, deep learning represents an unsupervised learning algorithm that learns representations of data through the use of neural nets.

Q10. What's the difference between a generative and dis- criminative model?

criminative Algorithm? (Stack Overflow)
A generative model will learn categories of data while a discriminative model will simply learn the distinction between different categories of data. Discriminative models will generally outperform generative models on classification tasks.

Q11. What cross-validation technique would you use on a time series dataset?

selection (CrossValidated)
Instead of using standard k-folds cross-validation, you have to pay attention to the fact that a time series is not randomly distributed data — it is inherently ordered by chronological order. If a pattern emerges in later time periods for example, your model may still pick up on it even if that effect doesn't hold in earlier years!
You'll want to do something like forward chaining where you'll be able to model on past data then look at forward-facing data.

- fold 1 : training [1], test [2]
- fold 2 : training [1 2], test [3]
- fold 3 : training [1 2 3], test [4]
- fold 4 : training [1 2 3 4], test [5]
- fold 5 : training [1 2 3 4 5], test [6]

Q12. How is a decision tree pruned?

Pruning is what happens in decision trees when branches that have weak predictive power are removed in order to reduce the complexity of the model and increase the predictive accuracy of a decision tree model. Pruning can happen bottom-up and top-down, with approaches such as reduced error pruning and cost complexity pruning.
Reduced error pruning is perhaps the simplest version: replace each node. If it doesn't decrease predictive accuracy, keep it pruned. While simple, this heuristic actually comes pretty close to an approach that would optimize for maximum accuracy.

Q13. Which is more important to you– model accuracy, or model performance?

This question tests your grasp of the nuances of machine learning model performance! Machine learning interview questions often look towards the details. There are models with higher accuracy that can perform worse in predictive power — how does that make sense? Well, it has everything to do with how model accuracy is only a subset of model performance, and at that, a sometimes misleading one. For example, if you wanted to detect fraud in a massive dataset with a sample of millions, a more accurate model would most likely predict no fraud at all if only a vast minority of cases were fraud. However, this would be useless for a predictive model — a model designed to find fraud that asserted there was no fraud at all! Questions like this help you demonstrate that you understand model accuracy isn't the be-all and end-all of model performance.

Q14. What's the F1 score? How would you use it?

The F1 score is a measure of a model's performance. It is a weighted average of the precision and recall of a model, with results tending to 1 being the best, and those tending to 0 being the worst. You would use it in classification tests where true negatives don't matter much.

Q15. How would you handle an imbalanced dataset?

Machine Learning Dataset (Machine Learning Mastery)
An imbalanced dataset is when you have, for example, a classification test and 90% of the data is in one class. That leads to problems: an accuracy of 90% can be skewed if you have no predictive power on the other category of data! Here are a few tactics to get over the hump:

1- Collect more data to even the imbalances in the dataset.
2- Resample the dataset to correct for imbalances.
3- Try a different algorithm altogether on your dataset.

What's important here is that you have a keen sense for what damage an unbalanced dataset can cause, and how to balance that.

Q16. When should you use classification over regression?

Classification produces discrete values and dataset to strict categories, while regression gives you continuous results that allow you to better distinguish differences between individual points. You would use classification over regression if you wanted your results to reflect the belongingness of data points in your dataset to certain explicit categories (ex: If you wanted to know whether a name was male or female rather than just how correlated they were with male and female names.)

Q17. How do you ensure you're not overfitting with a model?

This is a simple restatement of a fundamental problem in machine learning: the possibility of overfitting training data and carrying the noise of that data through to the test set, thereby providing inaccurate generalizations.
There are three main methods to avoid overfitting:

1- Keep the model simpler: reduce variance by taking into account fewer variables and parameters, thereby removing some of the noise in the training data.
2- Use cross-validation techniques such as k-folds cross-validation.
3- Use regularization techniques such as LASSO that penalize certain model parameters if they're likely to cause overfitting.

Q18. What evaluation approaches would you work to gauge the effectiveness of a machine learning model?

Learning Mastery)
You would first split the dataset into training and test sets, or perhaps use cross-validation techniques to further segment the dataset into composite sets of training and test sets within the data. You should then implement a choice selection of performance metrics: here is a fairly comprehensive list. You could use measures such as the F1 score, the accuracy, and the confusion matrix. What's

important here is to demonstrate that you understand the nuances of how a model is measured and how to choose the right performance measures for the right situations.

Q19. How would you evaluate a logistic regression model?

gistic Regression in Plain English

A subsection of the question above. You have to demonstrate an understanding of what the typical goals of a logistic regression are (classification, prediction, etc.) and bring up a few examples and use cases.

Q20. What's the "kernel trick" and how is it useful?

The Kernel trick involves kernel functions that can enable in higher-dimension spaces without explicitly calculating the coordinates of points within that dimension: instead, kernel functions compute the inner products between the images of all pairs of data in a feature space. This allows them the very useful attribute of calculating the coordinates of higher dimensions while being computationally cheaper than the explicit calculation of said coordinates. Many algorithms can be expressed in terms of inner products. Using the kernel trick enables us effectively run algorithms in a high-dimensional space with lower-dimensional data.

(Learn about Springboard's AI / Machine Learning Bootcamp, the first of its kind to come with a job guarantee.)

Machine Learning Interview Questions:
Programming

These machine learning interview questions test your knowledge of programming principles you need to implement machine learning principles in practice. Machine learning interview questions tend to be technical questions that test your logic and programming skills: this section focuses more on the latter.

Q21. How do you handle missing or corrupted data in a dataset?

You could find missing/corrupted data in a dataset and either drop those rows or columns, or decide to replace them with another value. In Pandas, there are two very useful methods: isnull() and dropna() that will help you find columns of data with missing or corrupted data and drop those values. If you want to fill the invalid values with a placeholder value (for example, 0), you could use the fillna() method.

Q22. Do you have experience with Spark or big data tools for machine learning?

You'll want to get familiar with the meaning of big data for different companies and the different tools they'll want. Spark is the big data tool most in demand now, able to handle immense datasets with speed. Be honest if you don't have experience with the tools demanded, but also take a look at job descriptions and see what tools pop up: you'll want to invest in familiarizing yourself with them.

Q23. What are some differences between a linked list and an array?

An array is an ordered collection of objects. A linked list is a series of objects with pointers that direct how to process them sequentially. An array assumes that every element has the same size, unlike the linked list. A linked list can more easily grow organically: an array has to be pre-defined or re-defined for organic growth. Shuffling a linked list involves changing which points direct where — meanwhile, shuffling an array is more complex and takes more memory.

Q24. Which data visualization libraries do you use? What are your thoughts on the best data visualization tools?

What's important here is to define your views on how to properly visualize data and your personal preferences when it comes to tools. Popular tools include R's ggplot, Python's seaborn and matplotlib, and tools such as Plot.ly and Tableau.

Related: 20 Python Interview Questions

Machine Learning Interview Questions:

Company/Industry Specific

These machine learning interview questions deal with how to implement your general machine learning knowledge to a specific company's requirements. You'll be asked to create case studies and extend your knowledge of the company and industry you're applying for with your machine learning skills.

Q25. How would you implement a recommendation system for our company's users?

Overflow)

A lot of machine learning interview questions of this type will involve implementation of machine learning models to a company's problems. You'll have to research the company and its industry in-depth, especially the revenue drivers the company has, and the types of users the company takes on in the context of the industry it's in.

Q26. How can we use your machine learning skills to gen- erate revenue?

This is a tricky question. The ideal answer would demonstrate knowledge of what drives the business and how your skills could relate. For example, if you were interviewing for music-streaming startup Spotify, you could remark that your skills at developing a better recommendation model would increase user retention, which would then increase revenue in the long run.

The startup metrics Slideshare linked above will help you understand exactly what performance indicators are important for startups and tech companies as they think about revenue and growth.

Q27. What do you think of our current data process?

This kind of question requires you to listen carefully and impart feedback in a manner that is constructive and insightful. Your interviewer is trying to gauge if you'd be a valuable member of their team and whether you grasp the nuances of why certain things are set the way they are in the company's data process based on company- or industry-specific conditions. They're trying to see if you can be an intellectual peer. Act accordingly.

Machine Learning Interview Questions: General

Machine Learning Interest

This series of machine learning interview questions attempts to gauge your passion and interest in machine learning. The right answers will serve as a testament for your commitment to being a lifelong learner in machine learning.

Q28. What are the last machine learning papers you've read?

machine learning?

Keeping up with the latest scientific literature on machine learning is a must if you want to demonstrate interest in a machine learning position. This overview of deep learning in Nature by the scions of deep learning themselves (from Hinton to Bengio to LeCun) can be a good reference paper and an overview of what's happening in deep learning — and the kind of paper you might want to cite.

Q29. Do you have research experience in machine learn- ing?

Related to the last point, most organizations hiring for machine learning positions will look for your formal experience in the field. Research papers, co-authored or supervised by leaders in the field, can make the difference between you being hired and not. Make sure you have a summary of your research experience and papers ready — and an explanation for your background and lack of formal research experience if you don't.

Q30. What are your favorite use cases of machine learning models?

learning algorithms? (Quora)

The Quora thread above contains some examples, such as decision trees that categorize people into different tiers of intelligence based on IQ scores. Make sure that you have a few examples in mind and describe what resonated with you. It's important that you demonstrate an interest in how machine learning is implemented.

Q31. How would you approach the "Netflix Prize" competi- tion?

The Netflix Prize was a famed competition where Netflix offered $1,000,000 for a better collaborative filtering algorithm. The team that won called BellKor had a 10% improvement and used an ensemble of different methods to win. Some familiarity with the case and its solution will help demonstrate you've paid attention to machine learning for a while.

Q32. Where do you usually source datasets?

Project (Springboard)

Machine learning interview questions like these try to get at the heart of your machine learning interest. Somebody who is truly passionate about machine learning will have gone off and done side projects on their own, and have a good idea of what great datasets are out there. If you're missing any, check out Quandl for economic and financial data, and Kaggle's Datasets collection for another great list.

Q33. How do you think Google is training data for self- driving cars?

Machine learning interview questions like this one really test your knowledge of different machine learning methods, and your inventiveness if you don't know the answer. Google is currently using recaptcha to source labeled data on storefronts and traffic signs. They are also building on training data collected by Sebastian Thrun at GoogleX — some of which was obtained by his grad students driving buggies on desert dunes!

Q34. How would you simulate the approach AlphaGo took to beat Lee Sidol at Go?

and tree search (Nature)
AlphaGo beating Lee Sidol, the best human player at Go, in a best-of-five series was a truly seminal event in the history of machine learning and deep learning. The Nature paper above describes how this was accomplished with "Monte-Carlo tree search with deep neural networks that have been trained by supervised learning, from human expert games, and by reinforcement learning from games of self-play."
Related: 40 Artificial Intelligence Interview Questions
Looking to land a role as a machine learning engineer?
Find out about Springboard's Machine Learning Engineering Career Track, the first of its kind to come with a job guarantee.

Q35. What is Machine learning?

Machine learning is a branch of computer science which deals with system programming in order to automatically learn and improve with experience. For example: Robots are programed so that they can perform the task based on data they gather from sensors. It automatically learns programs from data.

Q36. Mention the difference between Data Mining and Machine learning?

Machine learning relates with the study, design and development of the algorithms that give computers the capability to learn without being explicitly programmed. While, data mining can be defined as the process in which the unstructured data tries to extract knowledge or unknown interesting patterns. During this process machine, learning algorithms are used.

Q37. What is 'Overfitting' in Machine learning?

In machine learning, when a statistical model describes random error or noise instead of underlying relationship 'overfitting' occurs. When a model is excessively complex, overfitting is normally observed, because of having too many parameters with respect to the number of training data types. The model exhibits poor performance which has been overfit.

Q38. Why overfitting happens?

The possibility of overfitting exists as the criteria used for training the model is not the same as the criteria used to judge the e■cacy of a model.

Q39. How can you avoid overfitting ?

By using a lot of data overfitting can be avoided, overfitting happens relatively as you have a small dataset, and you try to learn from it. But if you have a small database and you are forced to come

with a model based on that. In such situation, you can use a technique known as cross validation. In this method the dataset splits into two section, testing and training datasets, the testing dataset will only test the model while, in training dataset, the datapoints will come up with the model.
In this technique, a model is usually given a dataset of a known data on which training (training data set) is run and a dataset of unknown data against which the model is tested. The idea of cross validation is to define a dataset to "test" the model in the training phase.

Q40. What is inductive machine learning?

The inductive machine learning involves the process of learning by examples, where a system, from a set of observed instances tries to induce a general rule.

Q41. What are the five popular algorithms of Machine Learning?

a) Decision Trees
b) Neural Networks (back propagation)
c) Probabilistic networks
d) Nearest Neighbor
e) Support vector machines

Q42. What are the different Algorithm techniques in Machine Learning?

The different types of techniques in Machine Learning are
a) Supervised Learning
b) Unsupervised Learning
c) Semi-supervised Learning
d) Reinforcement Learning
e) Transduction
f) Learning to Learn

Q43. What are the three stages to build the hypotheses or model in machine learning?

a) Model building
b) Model testing
c) Applying the model

Q44. What is the standard approach to supervised learning?

The standard approach to supervised learning is to split the set of example into the training set and the test.

Q45. What is 'Training set' and 'Test set'?

In various areas of information science like machine learning, a set of data is used to discover the potentially predictive relationship known as 'Training Set'. Training set is an examples given to the learner, while Test set is used to test the accuracy of the hypotheses generated by the learner, and it is the set of example held back from the learner. Training set are distinct from Test set.

Q46. List down various approaches for machine learning?

The different approaches in Machine Learning are
a) Concept Vs Classification Learning
b) Symbolic Vs Statistical Learning
c) Inductive Vs Analytical Learning

Q47. What is not Machine Learning?

a) Artificial Intelligence
b) Rule based inference

Q48. Explain what is the function of 'Unsupervised Learning'?

a) Find clusters of the data

b) Find low-dimensional representations of the data

c) Find interesting directions in data

d) Interesting coordinates and correlations

e) Find novel observations/ database cleaning

Q49. Explain what is the function of 'Supervised Learning'?

a) Classifications

b) Speech recognition

c) Regression

d) Predict time series

e) Annotate strings

Q50. What is algorithm independent machine learning?

Machine learning in where mathematical foundations is independent of any particular classifier or learning algorithm is referred as algorithm independent machine learning?

Q51. What is the difference between artificial learning and machine learning?

Designing and developing algorithms according to the behaviours based on empirical data are known as Machine Learning. While artificial intelligence in addition to machine learning, it also covers other aspects like knowledge representation, natural language processing, planning, robotics etc.

Q52. What is classifier in machine learning?

A classifier in a Machine Learning is a system that inputs a vector of discrete or continuous feature values and outputs a single discrete value, the class.

Q53. What are the advantages of Naive Bayes?

In Naïve Bayes classifier will converge quicker than discriminative models like logistic regression, so you need less training data. The main advantage is that it can't learn interactions between features.

Q54. In what areas Pattern Recognition is used?

Pattern Recognition can be used in

a) Computer Vision

b) Speech Recognition

c) Data Mining

d) Statistics

e) Informal Retrieval

f) Bio-Informatics

Q55. What is Genetic Programming?

Genetic programming is one of the two techniques used in machine learning. The model is based on the testing and selecting the best choice among a set of results.

Q56. What is Inductive Logic Programming in Machine Learning?

Inductive Logic Programming (ILP) is a subfield of machine learning which uses logical programming representing background knowledge and examples.

Q57. What is Model Selection in Machine Learning?

The process of selecting models among different mathematical models, which are used to describe the same data set is known as Model Selection. Model selection is applied to the fields of statistics, machine learning and data mining.

Q58. What are the two methods used for the calibration in Supervised Learning?

The two methods used for predicting good probabilities in Supervised Learning are

a) Platt Calibration

b) Isotonic Regression

These methods are designed for binary classification, and it is not trivial.

Q59. Which method is frequently used to prevent overfitting?

When there is su■cient data 'Isotonic Regression' is used to prevent an overfitting issue.

Q60. What is the difference between heuristic for rule learning and heuristics for decision trees?

The difference is that the heuristics for decision trees evaluate the average quality of a number of disjointed sets while rule learners only evaluate the quality of the set of instances that is covered with the candidate rule.

Q61. What is Perceptron in Machine Learning?

In Machine Learning, Perceptron is an algorithm for supervised classification of the input into one of several possible non-binary outputs.

Q62. Explain the two components of Bayesian logic program?

Bayesian logic program consists of two components. The first component is a logical one ; it consists of a set of Bayesian Clauses, which captures the qualitative structure of the domain. The second component is a quantitative one, it encodes the quantitative information about the domain.

Q63. What are Bayesian Networks (BN) ?

Bayesian Network is used to represent the graphical model for probability relationship among a set of variables .

Q64. Why instance based learning algorithm sometimes referred as Lazy learning algorithm?

Instance based learning algorithm is also referred as Lazy learning algorithm as they delay the induction or generalization process until classification is performed.

Q65. What are the two classification methods that SVM (Support Vector Machine) can handle?

a) Combining binary classifiers

b) Modifying binary to incorporate multiclass learning

Q66. What is ensemble learning?

To solve a particular computational program, multiple models such as classifiers or experts are strategically generated and combined. This process is known as ensemble learning.

Q67. Why ensemble learning is used?

Ensemble learning is used to improve the classification, prediction, function approximation etc of a model.

Q68. When to use ensemble learning?

Ensemble learning is used when you build component classifiers that are more accurate and independent from each other.

Q69. What are the two paradigms of ensemble methods?

The two paradigms of ensemble methods are

a) Sequential ensemble methods

b) Parallel ensemble methods

Q70. What is the general principle of an ensemble method and what is bagging and

boosting in ensemble method?

The general principle of an ensemble method is to combine the predictions of several models built with a given learning algorithm in order to improve robustness over a single model. Bagging is a method in ensemble for improving unstable estimation or classification schemes. While boosting method are used sequentially to reduce the bias of the combined model. Boosting and Bagging both can reduce errors by reducing the variance term.

Q71. What is bias-variance decomposition of classification error in ensemble method?

The expected error of a learning algorithm can be decomposed into bias and variance. A bias term measures how closely the average classifier produced by the learning algorithm matches the target function. The variance term measures how much the learning algorithm's prediction fluctuates for different training sets.

Q72. What is an Incremental Learning algorithm in ensemble?

Incremental learning method is the ability of an algorithm to learn from new data that may be available after classifier has already been generated from already available dataset.

Q73. What is PCA, KPCA and ICA used for?

PCA (Principal Components Analysis), KPCA (Kernel based Principal Component Analysis) and ICA (Independent Component Analysis) are important feature extraction techniques used for dimensionality reduction.

Q74. What is dimension reduction in Machine Learning?

In Machine Learning and statistics, dimension reduction is the process of reducing the number of random variables under considerations and can be divided into feature selection and feature extraction

Q75. What are support vector machines?

Support vector machines are supervised learning algorithms used for classification and regression analysis.

Q76. What are the components of relational evaluation techniques?

The important components of relational evaluation techniques are

a) Data Acquisition

b) Ground Truth Acquisition

c) Cross Validation Technique

d) Query Type

e) Scoring Metric

f) Significance Test

Q77. What are the different methods for Sequential Supervised Learning?

The different methods to solve Sequential Supervised Learning problems are

a) Sliding-window methods

b) Recurrent sliding windows

c) Hidden Markow models

d) Maximum entropy Markow models

e) Conditional random fields

f) Graph transformer networks

Q78. What are the areas in robotics and information processing where sequential
prediction problem arises?

The areas in robotics and information processing where sequential prediction problem arises are

a) Imitation Learning

b) Structured prediction

c) Model based reinforcement learning

Q79. What is batch statistical learning?
Statistical learning techniques allow learning a function or predictor from a set of observed data that can make predictions about unseen or future data. These techniques provide guarantees on the performance of the learned predictor on the future unseen data based on a statistical assumption on the data generating process.

Q80. What is PAC Learning?
PAC (Probably Approximately Correct) learning is a learning framework that has been introduced to analyze learning algorithms and their statistical e■ciency.

Q81. What are the different categories you can categorized the sequence learning
process?

a) Sequence prediction

b) Sequence generation

c) Sequence recognition

d) Sequential decision

Q82. What is sequence learning?
Sequence learning is a method of teaching and learning in a logical manner.

Q83. What are two techniques of Machine Learning ?
The two techniques of Machine Learning are

a) Genetic Programming

b) Inductive Learning

Q84. Give a popular application of machine learning that you see on day to day basis?
The recommendation engine implemented by major ecommerce websites uses Machine Learning

Q85. What is the Central Limit Theorem and why is it important?
Suppose that we are interested in estimating the average height among all people. Collecting data for every person in the world is impossible. While we can't obtain a height measurement from everyone in the population, we can still sample some people. The question now becomes, what can we say about the average height of the entire population given a single sample. The Central Limit Theorem addresses this

question exactly.

Q86. What is sampling? How many sampling methods do you know?

Data sampling is a statistical analysis technique used to select, manipulate and analyze a representative subset of data points to identify patterns and trends in the larger data set being examined." Read the full answer here.

Q87. What is the difference between type I vs type II error?

A type I error occurs when the null hypothesis is true, but is rejected. A type II error occurs when the null hypothesis is false, but erroneously fails to be rejected.

Q88. What is linear regression? What do the terms p-value, coefficient, and r-squared value mean? What is the significance of each of these components?

A linear regression is a good tool for quick predictive analysis: for example, the price of a house depends on a myriad of factors, such as its size or its location. In order to see the relationship between these variables, we need to build a linear regression, which predicts the line of best fit between them and can help conclude whether or not these two factors have a positive or negative relationship. Read more here and here.

Q89. What are the assumptions required for linear regression?

There are four major assumptions: 1. There is a linear relationship between the dependent variables and the regressors, meaning the model you are creating actually fits the data, 2. The errors or residuals of the data are normally distributed and independent from each other, 3. There is minimal multicollinearity between explanatory variables, and 4. Homoscedasticity. This means the variance around the regression line is the same for all values of the predictor variable.

Q90. What is a statistical interaction?

Basically, an interaction is when the effect of one factor (input variable) on the dependent variable (output variable) differs among levels of another factor.

Q91. What is selection bias?

Selection (or 'sampling') bias occurs in an 'active,' sense when the sample data that is gathered and prepared for modeling has characteristics that are not representative of the true, future population of cases the model will see. That is, active selection bias occurs when a subset of the data are systematically (i.e., non-randomly) excluded from analysis.

Q92. What is an example of a data set with a non-Gaussian distribution?

The Gaussian distribution is part of the Exponential family of distributions, but there are a lot more of them, with the same sort of ease of use, in many cases, and if the person doing the machine learning has a solid grounding in statistics, they can be utilized where appropriate.

Q93. What is the Binomial Probability Formula?

The binomial distribution consists of the probabilities of each of the possible numbers of successes on N trials for independent events that each have a probability of π (the Greek letter pi) of occurring." Read more

Data Science :

Q1. What is Data Science? List the differences between supervised and unsupervised
learning.

Data Science is a blend of various tools, algorithms, and machine learning principles with the goal to discover hidden patterns from the raw data. How is this different from what statisticians have been doing for years?

The answer lies in the difference between explaining and predicting.

The differences between supervised and unsupervised learning are as follows;

Supervised Learning Unsupervised Learning

Input data is labelled. Input data is unlabelled.
Uses a training data set. Uses the input data set.
Used for prediction. Used for analysis.
Enables classification and regression. Enables Classification, Density Estimation, & Dimension Reduction
Q2. What is Selection Bias?
Selection bias is a kind of error that occurs when the researcher decides who is going to be studied. It is usually associated with research where the selection of participants isn't random. It is sometimes referred to as the selection effect. It is the distortion of statistical analysis, resulting from the method of collecting samples. If the selection bias is not taken into account , then some conclusions of the study may not be accurate.
The types of selection bias include:

Q94. Attrition: Attrition bias is a kind of selection bias caused by attrition (loss of participants) discounting trial subjects/tests that did not run to completion. Q3. What is bias-variance trade-off?

Bias: Bias is an error introduced in your model due to oversimplification of the machine learning al gorithm. It can lead to underfitting. When you train your model at that time model makes simplified assumptions to make the target function easier to understand.
Low bias machine learning algorithms — Decision Trees, k -NN and SVM High bias machine learning algorithms — Linear Regression, Logistic Regression
Variance: Variance is error introduced in your model due to complex machine learning algorithm, your model learns noise also from the training data set and performs badly on test data set. It can lead to high sensitivity and overfitting.
Normally, as you increase the complexity of your model, you will see a reduction in error due to lower bias in the model. However, this only happens until a particular point. As you continue to make your model more complex, you end up over-fitting your model and hence your model will start suffering from high variance.
Bias-Variance trade-off: The goal of any supervised machine learning algorithm is to have low bias and low variance to achieve good prediction performance.

Q95. Asymptotic Q7. What is correlation and covariance in statistics?

Covariance and Cor relation are two mathematical concepts; these two approaches are widely used in statistics. Both Correlation and Covariance establish the relationship and also measure t he dependency between two random variables. Though the work is similar between these tw o in mathematical terms, they are different from each other.
Correlation:
Correlation is considered or described as the best technique for measuring and also for estima ting the quantitative relationship between two variables. Correlation measures how stro ngly two variables are related.
Covariance: In covariance two items vary together and it's a measure that indicates the extent to which two random variables change in cycle. It is a statistical term; it explains the systematic relation between a pair of random variables, wherein changes in one variable reciprocal by a corresponding change in another variable.
Q8. What is the difference between Point Estimates and Confidence Interval?
Point Estimation gives us a particular value as an estimate of a populatio n parameter. Method of Moments and Maximum Likelihood estimator methods are used to derive Point Estimators for population parameters.
A confidence interval gives us a r ange of values which is likely to contain the population parameter. The confidence interval is generally preferred, as it tells us how likely this interval is to contain the population parameter. This likeliness or probability is called Confidence Level or Confidence coefficient and represented by 1 — alpha, where alpha is the level of significance.

Q9. What is the goal of A/B Testing?

It is a hypothesis testing for a randomized experiment with two variables A and B.

The goal of A/B Testing is to identify any changes to the web page to maximize or increase the outcome of interest. A/B testing is a fantastic method for figuring out the best online promotional and marketing strategies for your business. It can be used to test everything from website copy to sales emails to search ads

An example of this could be identifying the click-through rate for a banner ad.

Q10. What is p-value?

When you perform a hypothesis test in statistics, a p -value can help you determine the strength of your results. p-value is a number between 0 and 1. Based on the value it will denote the strength of the results. The claim which is on trial is called the Null Hypothesis.

Low p -value (≤ 0.05) indicates strength against the null hypothesis which means we can reject the null Hypothesis. High p-value (≥ 0.05) indicates strength for the null hypothesis which means we can accept the null Hypothesis p-value of 0.05 indicates the Hypothesis could go either way. To put it in another way,

High P values: your data are likely with a true null. Low P values: your data are unlikely with a true null.

Q11. In any 15-minute interval, there is a 20% probability that you will see at least one shooting star. What is the probability that you see at least one shooting star in the period of an hour?

Probability of not seeing any shooting star in 15 minutes is

= 1 – P(Seeing one shooting s tar)

= 1 – 0.2 = 0.8

Probability of not seeing any shooting star in the period of one hour

= (0.8) ^ 4 = 0.4096

Probability of seeing at least one shooting star in the one hour

= 1 – P(Not seeing any star)

= 1 – 0.4096 = 0.5904

Q12. How can you generate a random number between 1 – 7 with only a die?

- Any die has six sides from 1-6. There is no way to get seven equal outcomes from a single rolling of a die. If we roll the die twice and consider the event of two rolls, we now have 36 different outcomes.
- To get our 7 equal outcomes we have to reduce this 36 to a number divisible by 7. We can thus consider only 35 outcomes and exclude the other one.
- A simple scenario can be to exclude the combination (6,6), i.e., to roll the die again if 6 appears twice.
- All the remaining combinations from (1,1) till (6,5) can be divided into 7 parts of 5 each. This way all the seven sets of outcomes are equally likely.

Q13. A certain couple tells you that they have two children, at least one of which is a girl. What is the probability that they have two girls?

In the case of two children, there are 4 equally likely possibilities

BB, BG, GB and GG;

where B = Boy and G = Girl and the first letter denotes the first child.

From the question, we can exclude the first case of BB. Thus from the remaining 3 possibilities of BG, GB & BB, we have to find the probability of the case with two girls.

Thus, P(Having two girls given one girl) = 1 / 3

Q14. A jar has 1000 coins, of which 999 are fair an d 1 is double headed. Pick a coin at random, and toss it 10 times. Given that you see 10 heads, what is the probability that the next toss of that coin is also a head?

There are two ways of choosing the coin. One is to pick a fair coin and the other is to pick the one with two heads.

Probability of sele cting fair coin = 999/1000 = 0.999

Probability of selecting unfair coin = 1/1000 = 0.001

Selecting 10 heads in a row = Selecting fair coin * Getting 10 heads + Selecting an unfair coin

P (A) = 0.999 * (1/ 2)^5 = 0.999 * (1/1024) = 0.000976

P (B) = 0.001 * 1 = 0.001

P(A / A + B) = 0.000976 / (0.000976 + 0.001) = 0.4939

P(B / A + B) = 0.001 / 0.001976 = 0.5061

Probability of selecting another head = P(A/A+B) * 0.5 + P(B/A+B) * 1 = 0.4939 * 0.5 + 0.5061 = 0.7531

Q15. What do you understand by statistical power of sensitivity and how do you calculate it?

Sensitivity is commonly used to validate the accuracy of a classifier (Logistic, SVM, Random Forest etc.).

Sensitivity is nothing but "Predicted True events/ Total events". True events here are the events which were true and model also predicted them as true.

Calculation of seasonality is pretty straightforward.

Seasonality = (True Positives) / (Positives in Actual Dependent Variable)

Q16. Why Is Re-sampling Done?

Resampling is done in any of these cases:

- Estimating the accuracy of sample statistics by using subsets of accessible data or drawing randomly with replacement from a set of data points
- Substituting labels on data points when performing significance tests
- Validating models by using random subsets (bootstrapping, cross-validation)

Q17. What are the differences between over-fitting and under-fitting?

In statistics and machine learning, one of the most common tasks is to fit a model to a set of training data, so as to be able to make reliable predictions on general untrained data.

In overfitting, a statistical model describes random error or noise instead of the underlying relationship. Overfitting occurs when a model is excessively complex, such as having too many parameters relative to the number of observations. A model that has been overfitted, has poor predictive performance, as it overreacts to minor fluctuations in the training data.

Underfitting occurs when a statistical mode l or machine learning algorithm ca nnot capture the underlying trend of the data. Underfitting would occur, for example, when fitting a linear model to non-linear data. Such a model too would have poor predictive performance.

Q18. How to combat Overfitting and Underfitting?

To combat overfi tting and underfitting, you can resample the data to estimate the model accuracy (k -fold cross-validation) and by having a validation dataset to evaluate the model.

Q19. What is regularisation? Why is it useful?

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Regularisation is the process of adding tuning parameter to a model to induce smoothness in order to prevent overfitting. This is most often done by adding a constant multiple to an existing weight vector. This constant is often the L1(Lasso) or L2(ridge). The model predictions should then minimize the loss function calculated on the regularized training set.

Q20. What Is the Law of Large Numbers?

It is a theorem that describes the result of performing the same experiment a large number of times. This theorem forms the b asis of frequency-style thinking. It says that the sample means, the sample variance and the sample standard deviation converge to what they are trying to estimate.

Q21. What Are Confounding Variables?

In statistics, a confounder is a variable that influences both the dependent variable and independent variable.

For example, if you are researching whether a lack of exercise leads to weight gain,

lack of exercise = independent variable

weight gain = dependent variable.

A confounding variable here would be any other variable that affects both of these variables, such as the age of the subject.

Q22. What Are the Types of Biases That Can Occur During Sampling?

- Selection bias
- Under coverage bias
- Survivorship bias

Q23. What is Survivorship Bias?

It is the logical error of focusing aspects that support surviving some process and casually overlooking those that did not work becaus e of their lack of prominence. This can lead to wrong conclusions in numerous different means.

Q24. What is selection Bias?

Selection bias occurs when the sample obtained is not representative of the population intended to be analysed.

Q25. Explain how a ROC curve works?

The ROC curve is a graphical representation of the contrast between true positive rates and false -positive rates at various thresholds. It is often used as a proxy for the trade -off between the sensitivity(true positive rate) and false-positive rate.

Q26. What is TF/IDF vectorization?

TF–IDF is short for term frequency -inverse document frequency, is a numerical statistic that is intended to reflect how important a word is to a document in a collection or corpus. It is often used as a weighting factor in information retrieval and text mining.

The TF–IDF value increases proportionally to the number of times a word appears in the document but is offset by the frequency of the word in the corpus, which helps to adjust for the fact that some words appear more frequently in general.

Q27. Why we generally use Softmax non-linearity function as last operation in-network?

It is because it takes in a vector of real numbers and returns a probability distribution. Its definition is as follows. Let x be a vector of real numbers (positive, negative, whatever, there are no constraints).

Then the i'th component of Softmax(x) is —

It should be clear that the output is a probability distribution: each element is non-negative and the sum over all components is 1.

DATA ANALYSIS INTERVIEW QUESTIONS

Q28. Python or R – Which one would you prefer for text analytics?

We will prefer Python because of the following reasons:

- Python would be the best option because it has Pandas library that provides easy to use data structures and high-performance data analysis tools.
- R is more suitable for machine learning than just text analysis.
- Python performs faster for all types of text analytics.

Q29. How does data cleaning plays a vital role in the analysis?

Data cleaning can help in analysis because:

- Cleaning data from multiple sources helps to transform it in to a format that data analysts or data scientists can work with.
- Data Cleaning helps to increase the accuracy of the model in machine learning.
- It is a cumbersome process because as the number of data sources increases, the time taken to clean the data inc reases exponentially due to the number of sources and the volume of data generated by these sources.
- It might take up to 80% of the time for just cleaning data making it a critical part of the analysis task.

Q30. Differentiate between univariate, bivariate and multivariate analysis.

Univariate analyses are descriptive statistical analysis techniques which can be differentiated based on the number of variables involved at a given point of time. For example, the pie charts of sales based on territory involve only one variable and can the analysis can be referred to as univariate analysis.

The bivariate analysis attempts to understand the difference between two variables at a time as in a scatterplot. For example, analyzing the volume of sale and spending can b e considered as an example of bivariate analysis.

Multivariate analysis deals with the study of more than two variables to understand the effect of variables on the responses.

Q31. Explain Star Schema.

It is a traditional database schema with a central tab le. Satellite tables map IDs to physical names or descriptions and can be connected to the central fact table using the ID fields; these tables are known as lookup tables and are principally useful in real -time applications, as they save a lot of memory. S ometimes star schemas involve several layers of summarization to recover information faster.

Q32. What is Cluster Sampling?

Cluster sampling is a technique used when it becomes difficult to study the target population spread across a wide area and simple random sampling cannot be applied. Cluster Sample is a probability sample where each sampling unit is a collection or cluster of elements.

For eg., A researcher wants to survey the academic performance of high school students in Japan. He can divide the entire population of Japan into different clusters (cities). Then the researcher selects a number of clusters depending on his research through simple or systematic random sampling.

Let's continue our Data Science Interview Questions blog with some more statistics questions.

Q33. What is Systematic Sampling?

Systematic sampling is a statistical technique where elements are selected from an ordered sampling frame. In systematic sampling, the list is progressed in a circular manner so once you reach the end of the list, it is progressed from the top again. The best example of systematic sampling is equal probability method.

Q34. What are Eigenvectors and Eigenvalues?

Eigenvectors are used for understanding linear transformations. In data analysis, we usually calc ulate the eigenvectors for a correlation or covariance matrix. Eigenvectors are the directions along which a particular linear transformation acts by flipping, compressing or stretching.

Eigenvalue can be referred to as the strength of the transformation i n the direction of eigenvector or the factor by which the compression occurs.

Q35. Can you cite some examples where a false positive is important than a false negative?

Let us first understand what false positives and false negatives are.

- False Positives are the cases where you wrongly classified a non -event as an event a.k.a Type I error.
- False Negatives are the cases where you wrongly classify events as non-events, a.k.a Type II error.

Example 1: In the medical field, assume you have to give chemotherapy to patients. Assume a patient comes to that hospital and he is tested positive for cancer, based on the lab prediction but he actually doesn't have cancer. This is a case of false positive. Here it is of utmost danger to start chemotherapy on this patient when he actually does not have cancer. In the absence of cancerous cell, chemotherapy will do certain damage to his normal healthy cells and might lead to severe diseases, even cancer.

Example 2: Let's say an e-commerce company decided to give $1000 Gift voucher to the customers whom they assume to purchase at least $10,000 worth of items. They send free voucher mail directly to 100 customers without any minimum purchase condition because they assume to make at least 20% profit on sold items above $10,000. Now the issue is if we send the $1000 gift vouchers to customers who have not actually purchased anything but are marked as having made $10,000 worth of purchase.

Q36. Can you cite some examples where a false negative important than a false positive?

Example 1: Assume there is an airport 'A' which has received high -security threats and based on certain characteristics they identify whether a particular passenger can be a threat or not. Due to a shortage of staff, they decide to scan passengers being predicted as risk positives by their predictive model. What will happen if a true threat customer is being flagged as non-threat by airport model?

Example 2: What if Jury or judge decides to make a criminal go free?

Example 3: What if you rejec ted to marry a very good person based on your predictive model and you happen to meet him/her after a few years and realize that you had a false negative?

Q37. Can you cite some examples where both false positive and false negatives are equally important?

In the Banking industry giving loans is the primary source of making money but at the same time if your repayment rate is not good you will not make any profit, rather you will risk huge losses.

Banks don't want to lose good customers and at the same point in time, they don' t want to acquire bad customers. In this scenario, both the false positives and false negatives become very important to measure.

Q38. Can you explain the difference between a Validation Set and a Test Set?

A Validation set can be considered as a part of the training set as it is used for parameter selection and to avoid overfitting of the model being built.

On the other hand, a Test Set is used for testing or evaluating the performance of a trained machine learning model.

In simple terms, the differences can be summarized as; training set is to fit the parameters i.e. weights and test set is to assess the performance of the model i.e. evaluating the predictive power and generalization.

Q39. Explain cross-validation.

Cross-validation is a model validation te chnique for evaluating how the outcomes of statistical analysis will generalize to an independent dataset. Mainly used in backgrounds where the objective is forecast and one wants to estimate how accurately a model will accomplish in practice.

The goal of cross-validation is to term a data set to test the model in the training phase (i.e. validation data set) in order to limit problems like overfitting and get an insight on how the model will generalize to an independent data set.

MACHINE LEARNING INTERVIEW QUESTIONS

Q40. What is Machine Learning?

Machine Learning explores the study and construction of algorithms that can learn from and make predictions o n data. Closely rela ted to computational statistics. Used to devise complex models and algorithms that lend themselves to a prediction which in commercial use is known as predictive analytics.

Given below, is an image representing the various domains Machine Learning lends itself to.

Q41. What is Supervised Learning?

Supervised learning is the machine learning task of inferring a function from labeled training data. The training data consist of a set of training examples.

Algorithms: Support Vector Machines, Regression, Naive Bayes, Decision Trees, K -nearest Neighbor Algorithm and Neural Networks

E.g. If you built a fruit classifier, the labels will be "this is an orange, this is an apple and this i s a banana", based on showing the classifier examples of apples, oranges and bananas.

Q42. What is Unsupervised learning?

Unsupervised learnin g is a type of machine learning alg orithm used to draw inferences from datasets consisting of input data without labelled responses.

Algorithms: Clustering, Anomaly Detection, Neural Networks and Latent Variable Models

E.g. In the same example, a fruit clustering will categorize as "fruits with soft skin and lots of dimples", "fruits with shiny hard skin" and "elongated yellow fruits".

Q43. What are the various classification algorithms?

The diagram lists the most important classification algorithms.

Q44. What is 'Naive' in a Naive Bayes?
The Naive Bayes Algorithm is based on the Bayes Theorem. Bayes' theorem describes the probability o f an event, based on prior knowledge of conditions that might be related to the event.
The Algorithm is 'naive' because it makes assumptions that may or may not turn out to be correct.
Q45. Explain SVM algorithm in detail.
SVM stands for support vector machine, it is a supervised machine learning algorithm which can be used for both Regression and Classification. If you have n features in your training data set, SVM tries to plot it in n-dimensional space with the value of each feature being the value of a particular coordinate. SVM uses hyperplanes to separate out different classes based on the provided kernel function.
Q46. What are the support vectors in SVM?
In the diagram, we see that the thinner lines mark the distance from the classifier to the closest data points called the support vectors (darkened data points). The distance between t he two thin lines is called the margin.
Q47. What are the different kernels in SVM?
There are four types of kernels in SVM.

Q96. To just remove the value. Q58. What are the various steps involved in an analytics project?

The following are the various steps involved in an analytics project:

Q97. Start implementing the model and track the result to analyze the performance of the model over the period of time. Q59. During analysis, how do you treat missing values?

The extent of the missing values is identified after identifying the varia bles with miss ing values. If any patterns are identified the analyst has to concentrate on them as it could lead to interesting and meaningful business insights.
If there are no patterns identified, then the missing values can be substituted with mean or m edian values (imputation) or they can simply be ignored. Assigning a default value which can be mean, minimum or maximum value. Getting into the data is important.
If it is a categorical variable, the default value is assigned. The missing value is assigne d a default value. If you have a distribution of data coming, for normal distribution give the mean value.
If 80% of the values for a variable are missing then you can answer that you would be dropping the variable instead of treating the missing values.
Q60. How will you define the number of clusters in a clustering algorithm?
Though the Clustering Algorithm is not specified, this question is mostly in reference to K-Means clustering where "K" defines the number of clusters. The objective of clustering is to group similar entities in a way that the entities within a group are similar to each other but the groups are different from each other.
For example, the following image shows three different groups.
Within Sum of
squares is generally used to explain the homogeneity within a cluster. If you plot WSS for a range of number of clusters, you will get the plot shown below.

- The Graph is generally known as Elbow Curve.
- Red circled a point in above graph i.e. Number of Cluster =6 is the point after which you don't see

any decrement in WSS.

- This point is known as the bending point and taken as K in K – Means.

This is the widely used approach but few data scientists also use Hierarchical clustering f irst to create dendrograms and identify the distinct groups from there.
Q61. What is Ensemble Learning?
Ensemble Learning is basically combining a diverse set of learners(Individual models) together to improvise on the stability and predictive power of the model.
Q62. Describe in brief any type of Ensemble Learning?

Ensemble learning has many types but two more popular ensemble learning techniques are mentioned below.

Bagging

Bagging tries to implement similar learners on small sample populations and then t akes a mean of all the predictions. In generalised bagging, you can use different learners on different population. As you expect this helps us to reduce the variance error.

Boosting

Boosting is an iterative technique which adjusts the weight of an observation base d on the last classification. If an observation was classified incorrectly, it tries to increase the weight of this observation and vice versa. Boosting in general decreases the bias error and builds strong predictive models. However, they may over fit on the training data.

Q63. What is a Random Forest? How does it work?

Random forest is a versatile machine learning method capable of performing both regression and classification tasks. It is also used for dimensionality reduction, treats missing values, outlier values. It is a type of ensemble learning method, where a group of weak models combine to form a powerful model.

In Random Forest, we grow multiple trees as opposed to

a single tree. To classify a new object based on attributes, each tree gives a classification. The forest chooses the classification having the most votes(Overall the trees in the forest) and in case of regression, it takes the average of outputs by different trees.

Q64. How Do You Work Towards a Random Forest?

The underlying principle of this technique is that several weak learners combined to provide a keen learner. The steps involved are

- Build several decision trees on bootstrapped training samples of data
- On each tree, each time a split is considered, a random sample of mm predictors is chosen as split candidates, out of all pp predictors
- Rule of thumb: At each split m=p√m=p
- Predictions: At the majority rule

Q65. What cross-validation technique would you use on a time series data set?

Instead of using k-fold cross-validation, you should be aware of the fact that a time series is not randomly distributed data — It is inherently ordered by chronological order.

In case of time series data, you should use techniques like forward=chaining — Where you will be model on past data then look at forward-facing data.

fold 1: training[1], test[2]

fold 1: training[1 2], test[3]

fold 1: training[1 2 3], test[4]

fold 1: training[1 2 3 4], test[5]

Q66. What is a Box-Cox Transformation?

The dependent variable for a regression analysis might not satisfy one or more assumptions of an ordinary least squares regression. The residuals could either curve as the prediction increases or follow the skewed distribution. In such scenarios, it is necessary to transform the response variable so that the data meets the required assumptions. A Box cox transformation is a statistical technique to transform non-normal dependent variables into a normal shape. If the given data is not normal then most of the statistical techniques assume normality. Applying a box cox transformation means that you can run a broader number of tests.

A Box-Cox transformation is a way to transform non-normal dependent variables into a normal shape. Normality is an important assumption for many statistical techniques, if your data isn't normal, applying a Box-Cox means that you are able to run a broader number of tests. The Box-Cox transformation is named after statisticians George Box and Sir David Roxbee Cox who collaborated on a 1964 paper and developed the technique.

Q67. How Regularly Must an Algorithm be Updated?

You will want to update an algorithm when:

- You want the model to evolve as data streams through infrastructure
- The underlying data source is changing
- There is a case of non-stationarity
- The algorithm underperforms/ results lack accuracy

Q68. If you are having 4GB RAM in your machine and you want to train your model on 10GB data set. How would you go about this problem? Have you ever faced this kind of problem in your machine learning/data science experience so far?

First of all, you have to ask which ML model you want to train.

For Neural networks: Batch size with Numpy array will work.

Steps:

Q98. It gives better accuracy to the model since every neuron performs different computations. This is the most commonly used method. Q76. What Is the Cost Function?

Also referred to as "loss" or "error," cost function is a measur e to evaluate how good your model's performance is. It's used to compute the error of the output layer during backpropagation. We push that error backwards through the neural network and use that during the different training functions.

Q77. What Are Hyperparameters?

With neural networks, you're usually working with hyperparameters once the data is formatted correctly. A hyperparameter is a parameter whose value is set before the learning process begins. It determines how a network is trained and the struct ure of the network (such as the number of hidden units, the learning rate, epochs, etc.).

Q78. What Will Happen If the Learning Rate Is Set inaccurately (Too Low or Too High)?

When your learning rate is too low, training of the model will progress very slowly as we are making minimal updates to the weights. It will take many updates before reaching the minimum point.

If the learning rate is set too high, this causes unde sirable divergent behaviour to the loss function due to drastic updates in weights. It m ay fail to converge (model can give a good output) or even diverge (data is too chaotic for the network to train).

Q79. What Is the Difference Between Epoch, Batch, and Iteration in Deep Learning?

- Epoch – Represents one iteration over the entire dataset (everything put into the training model).
- Batch – Refers to when we cannot pass the entire dataset into the neural network at once, so we divide the dataset into several batches.
- Iteration – if we have 10,000 images as data and a batch size of 200. then an e poch should run 50 iterations (10,000 divided by 50).

Q80. What Are the Different Layers on CNN?

There are four layers in CNN:

Q99. Fully Connected Layer – this layer recognizes and classifies the objects in the image. Q81. What Is Pooling on CNN, and How Does It Work?

Pooling is used to reduce the spatial dimensions of a CNN. It performs down-sampling operations to reduce the dimensionality and creates a pooled feature map by sliding a filter matrix over the input matrix.

Q82. What are Recurrent Neural Networks(RNNs)?

RNNs are a type of artificial neural networks designed to recognise the pattern from the sequence of data such as Time series, stock market and government agencies etc. To understand recurrent nets, first, you have to understand the basics of feedforward nets.

Both these networks RNN and feed-forward named after the way they channel information through a series of mathematical orations performed at the nodes of the network. One feeds information through straight(never touching the same node twice), while the oth er cycles it through a loop, and the latter are called recurrent.

Recurrent networks, on the other hand, take as their input, not just the current input example they see, but also the what they have perceived previously in time.

The decision a recurrent neural network reached at time t-1 affects the decision that it will reach one moment later at time t. So recurrent networks have two sources of input, the present and the recent past, which combine to determine how they respond to new data, much as we do in life.

The error they generate will return via backpropagation and be used to adjust their weights until error can't go any lower. Remember, the purpose of recurrent nets is to accurately classify sequential input. We rely on the backpropagation of error and gradient descent to do so.

Q83. How Does an LSTM Network Work?

Long-Short-Term Memory (LSTM) is a special kind of recurrent neural network capable of learning long-term dependencies, remembering information for long periods as its default behaviour. There are three steps in an LSTM network:

- Step 1: The network decides what to forget and what to remember.
- Step 2: It selectively updates cell state values.
- Step 3: The network decides what part of the current state makes it to the output.

Q84. What Is a Multi-layer Perceptron(MLP)?

As in Neural Networks, MLPs have an input layer, a hidden layer, and an output layer. It has the same structure as a single layer perceptron with one or more hidden layers. A single layer perceptron can classify only linear separable classes with binary output (0,1), but MLP can classify nonlinear classes.

Except for the input layer, each node in the other
layers uses a nonlinear activation function. This means the input layers, the data coming in, and the activation function is based upon all nodes and weights being added together, producing the output. MLP uses a supervised learning method called "backpropagation." In backpropagation, the neural network calculates the error with the help of cost function. It propagates this error backward from where it came (adjusts the weights to train the model more accurately).

Q85. Explain Gradient Descent.

To Understand Gradient Descent, Let's understand what is a Gradient first.

A gradient measures how much the output of a function changes if you change the inputs a little bit. It simply measures the change in all weights with regard to the change in error. You can also think of a gradient as the slope of a function.

Gradient Descent can be thought of climbing down to the bottom of a valley, instead of climbing up a hill. This is because it is a minimization algorithm that minimizes a given function (Activation Function).

Q86. What is
exploding gradients?

While training an RNN, if you see exponentially growing (very large) error gradients which accumulate and result in very large updates to neural network model weights during training, they're known as exploding gradients. At an extreme, the values of weights can become so large as to overflow and result in NaN values.

This has the effect of your model is unstable and unable to learn from your training data.

Q87. What is vanishing gradients?

While training an RNN, your slope can become either too small; this makes the training difficult. When the slope is too small, the problem is known as a Vanishing Gradient. It leads to long training times, poor performance, and low accuracy.

Q89. What is Back Propagation and Explain it's Working.

Backpropagation is a training algorithm used for multilayer neural network. In this method, we move the error from an end of the network to all weights inside the network and thus allowing efficient computation of the gradient.

It has the following steps:

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Weekday / Weekend BatchesSee Batch Details

- Forward Propagation of Training Data
- Derivatives are computed using output and target
- Back Propagate for computing derivative of error wrt output activation
- Using previously calculated derivatives for output
- Update the Weights

Q90. What are the variants of Back Propagation?

- Stochastic Gradient Descent: We use only a single training example for calculation of gradient and update parameters.
- Batch Gradient Descent: We calculate the gradient for the whole dataset and perform the update at each iteration.
- Mini-batch Gradient Descent: It's one of the most popular optimization algorithms. It's a variant of Stochastic Gradient Descent and here instead of single training example, mini -batch of samples is used.

Q91. What are the different Deep Learning Frameworks?

- Pytorch
- TensorFlow
- Microsoft Cognitive Toolkit
- Keras
- Caffe
- Chainer

Q92. What is the role of the Activation Function?

The Activation function is used to introduce non-linearity into the neural netwo rk helping it to learn more complex function. Without which the neural network would be only able to learn linear function which is a linear combination of its input data. An activation function is a function in an artificial neuron that delivers an output based on inputs.

Q93. Name a few Machine Learning libraries for various purposes.

Purpose Libraries

Scientific Computation Numpy

Tabular Data Pandas

Data Modelling & Preprocessing Scikit Learn

Time-Series Analysis Statsmodels

Text processing Regular Expressions, NLTK

Deep Learning Tensorflow, Pytorch

Q94. What is an Auto-Encoder?

Auto-encoders are simple learning networks that aim to transform inputs into outputs with the minimum possible error. This means that we want the output to be as close to input as possible. We add a couple of layers between the input and the output, and the sizes of these layers are smaller than the input layer. The auto-encoder receives unlabelled input which is then encoded to reconstruct the input.

Q95. What is a Boltzmann Machine?

Boltzmann machines have a simple learning algorithm that allows them to discover interesting features that represent complex regularities in the training data. The Boltzmann machine is basically used to optimise the weights and the quantity for the g iven problem. The learning algorithm is very slow in networks with many layers of fea ture detectors. " Restricted Boltzmann Machines " algorithm has a single layer of feature detectors which makes it faster than the rest.

Q96. What Is Dropout and Batch Normalization?

Dropout is a technique of dropping out hidden and visible units of a network randomly to prevent overfitting of data (typically dropping 20 per cent of the nodes). It doubles the number of iterations needed to converge the network.

Batch normalization is the technique to improve the performance and stability of neural networks by normalizing the inputs in every layer so that they have mean output activation of zero and standard deviation of one.

Q97. What Is the Difference Between Bat ch Gradient Descent and Stochastic Gradient Descent?

Batch Gradient Descent Stochastic Gradient Descent

The batch gradient computes the gradient using the entire dataset.

The stochastic gradient computes the gradient using a single sample.

It takes time to converge because the volume of data is huge, and weights update slowly.

It converges much faster than the batch gradient because it updates weight more frequently.

Q98. Why Is Tensorflow the Most Preferred Library in Deep Learning?

Tensorflow provides both C++ and Python APIs, making it easier to work on and has a faster compilation time compared to other Deep Learning libraries like Keras and Torch. Tensorflow supports both CPU and GPU computing devices.

Q99. What Do You Mean by Tensor in Tensorflow?

A tensor is a mathematical object represented as arrays of higher dimensions. These arrays of data with different dimensions and ranks fed as input to the neural network are called "Tensors."

Q100. What is the Computational Graph?

Everything in a tensorflow is based on creating a computational graph. It has a network of nodes where each node operates, Nodes represent mathematical operations, and edges represent tensors. Since data fl ows in the form of a graph, it is also called a "DataFlow Graph."

Q101. What are the differences between supervised and unsupervised learning?

Supervised Learning Unsupervised Learning

- Uses known and labeled data as input
- Supervised learning has a feedback mechanism
- Most commonly used supervised learning algorithms are decision trees, logistic regression, and support vector machine
- Uses unlabeled data as input
- Unsupervised learning has no feedback mechanism
- Most commonly used unsupervised learning algorithms are k-means clustering, hierarchical clustering, and apriori algorithm

Q100. How is logistic regression done?

Logistic regression measures the relationship between the dependent variable (our label of what we want to predict) and one or more independent variables (our features) by estimating probability using its underlying logistic function (sigmoid).

The image shown below depicts how logistic regression works:

The formula and graph for the sigmoid function are as shown:

Q101. How do you build a random forest model?

A random forest is built up of a number of decision trees. If you split the data into different packages and make a decision tree in each of the different groups of data, the random forest brings all those trees together.

Steps to build a random forest model:

Q102. How can you avoid the overfitting your model?

Overfitting refers to a model that is only set for a very small amount of data and ignores the bigger picture.

There are three main methods to avoid overfitting:

Q103. What are the feature selection methods used to select the right variables?

There are two main methods for feature selection:

Filter Methods

This involves:

- Linear discrimination analysis
- ANOVA
- Chi-Square

The best analogy for selecting features is "bad data in, bad answer out." When we're limiting or selecting the features, it's all about cleaning up the data coming in.

Wrapper Methods

This involves:

- Forward Selection: We test one feature at a time and keep adding them until we get a good fit
- Backward Selection: We test all the features and start removing them to see what works better
- Recursive Feature Elimination: Recursively looks through all the different features and how they pair together

Wrapper methods are very labor-intensive, and high-end computers are needed if a lot of data analysis is performed with the wrapper method.

Q104. You are given a data set consisting of variables with more than 30 percent missing values. How will you deal with them?

The following are ways to handle missing data values:

If the data set is large, we can just simply remove the rows with missing data values. It is the quickest way; we use the rest of the data to predict the values.

For smaller data sets, we can substitute missing values with the mean or average of the rest of the data using pandas data frame in python. There are different ways to do so, such as df.mean(), df.fillna(mean).

Q105. For the given points, how will you calculate the Euclidean distance in Python?

plot1 = [1,3]

plot2 = [2,5]

The Euclidean distance can be calculated as follows:

euclidean_distance = sqrt((plot1[0]-plot2[0])**2 + (plot1[1]-plot2[1])**2)

Q106. What are dimensionality reduction and its benefits?

Dimensionality reduction refers to the process of converting a data set with vast dimensions into data with fewer dimensions (fields) to convey similar information concisely.

This reduction helps in compressing data and reducing storage space. It also reduces computation time as fewer dimensions lead to less computing. It removes redundant features; for example, there's no point in storing a value in two different units (meters and inches).

Q107. How will you calculate eigenvalues and eigenvectors of the following 3x3 matrix?

- 2 -4 2
- 2 1 2

4 2 5

The characteristic equation is as shown:

Expanding determinant:

$(-2 - \lambda) [(1-\lambda) (5-\lambda)-2x2] + 4[(-2) x (5-\lambda) -4x2] + 2[(-2) x 2-4(1-\lambda)] =0$

- $\lambda3 + 4\lambda2 + 27\lambda - 90 = 0,$

$\lambda3 - 4 \lambda2 -27 \lambda + 90 = 0$

Here we have an algebraic equation built from the eigenvectors.

By hit and trial:

$33 - 4 x 32 - 27 x 3 +90 = 0$

Hence, $(\lambda - 3)$ is a factor:

$\lambda3 - 4 \lambda2 - 27 \lambda +90 = (\lambda - 3) (\lambda2 - \lambda - 30)$

Eigenvalues are 3,-5,6:

$(\lambda - 3) (\lambda2 - \lambda - 30) = (\lambda - 3) (\lambda+5) (\lambda-6),$

Calculate eigenvector for $\lambda = 3$

For X = 1,

- 5 - 4Y + 2Z =0,
- 2 - 2Y + 2Z =0

Subtracting the two equations:

3 + 2Y = 0,

Subtracting back into second equation:

Y = -(3/2)

Z = -(1/2)

Similarly, we can calculate the eigenvectors for -5 and 6.

Q108. How should you maintain a deployed model?

The steps to maintain a deployed model are:

Monitor

Constant monitoring of all models is needed to determine their performance accuracy. When you change something, you want to figure out how your changes are going to affect things. This needs to be monitored to ensure it's doing what it's supposed to do.

Evaluate

Evaluation metrics of the current model are calculated to determine if a new algorithm is needed.

Compare

The new models are compared to each other to determine which model performs the best.

Rebuild

The best performing model is re-built on the current state of data.

Q109. What are recommender systems?

A recommender system predicts what a user would rate a specific product based on their preferences. It can be split into two different areas:

Collaborative filtering

As an example, Last.fm recommends tracks that other users with similar interests play often. This is also commonly seen on Amazon after making a purchase; customers may notice the following message accompanied by product recommendations: "Users who bought this also bought…"

Content-based filtering

As an example: Pandora uses the properties of a song to recommend music with similar properties. Here, we look at content, instead of looking at who else is listening to music.

Q110. How do you find RMSE and MSE in a linear regression model?

RMSE and MSE are two of the most common measures of accuracy for a linear regression model.

RMSE indicates the Root Mean Square Error.

MSE indicates the Mean Square Error.

Q111. How can you select k for k-means?

We use the elbow method to select k for k-means clustering. The idea of the elbow method is to run k-means clustering on the data set where 'k' is the number of clusters.

Within the sum of squares (WSS), it is defined as the sum of the squared distance between each member of the cluster and its centroid.

Q112. What is the significance of p-value?

p-value typically ≤ 0.05

This indicates strong evidence against the null hypothesis; so you reject the null hypothesis.

p-value typically > 0.05

This indicates weak evidence against the null hypothesis, so you accept the null hypothesis.

p-value at cutoff 0.05

This is considered to be marginal, meaning it could go either way.

Q113. How can outlier values be treated?

You can drop outliers only if it is a garbage value.

Example: height of an adult = abc ft. This cannot be true, as the height cannot be a string value. In this case, outliers can be removed.

If the outliers have extreme values, they can be removed. For example, if all the data points are clustered between zero to 10, but one point lies at 100, then we can remove this point.

If you cannot drop outliers, you can try the following:

- Try a different model. Data detected as outliers by linear models can be fit by nonlinear models.

Therefore, be sure you are choosing the correct model.

- Try normalizing the data. This way, the extreme data points are pulled to a similar range.
- You can use algorithms that are less affected by outliers; an example would be random forests.

Q114. How can a time-series data be declared as stationery?

It is stationary when the variance and mean of the series are constant with time.

Here is a visual example:

In the first graph, the variance is constant with time. Here, X is the time factor and Y is the variable. The value of Y goes through the same points all the time; in other words, it is stationary.

In the second graph, the waves get bigger, which means it is non-stationary and the variance is changing with time.

Q115. How can you calculate accuracy using a confusion matrix?

Consider this confusion matrix:

You can see the values for total data, actual values, and predicted values.

The formula for accuracy is:

Accuracy = (True Positive + True Negative) / Total Observations

= (262 + 347) / 650

= 609 / 650

= 0.93

As a result, we get an accuracy of 93 percent.

Q116. 'People who bought this also bought…' recommendations seen on Amazon are a result of which algorithm?

The recommendation engine is accomplished with collaborative filtering. Collaborative filtering explains the behavior of other users and their purchase history in terms of ratings, selection, etc.

The engine makes predictions on what might interest a person based on the preferences of other users. In this algorithm, item features are unknown.

For example, a sales page shows that a certain number of people buy a new phone and also buy tempered glass at the same time. Next time, when a person buys a phone, he or she may see a recommendation to buy tempered glass as well.

Q117. What is a Generative Adversarial Network?

Suppose there is a wine shop purchasing wine from dealers, which they resell later. But some dealers sell fake wine. In this case, the shop owner should be able to distinguish between fake and authentic wine.

The forger will try different techniques to sell fake wine and make sure specific techniques go past the shop owner's check. The shop owner would probably get some feedback from wine experts that some of the wine is not original. The owner would have to improve how he determines whether a wine is fake or authentic.

The forger's goal is to create wines that are indistinguishable from the authentic ones while the shop owner intends to tell if the wine is real or not accurately

Let us understand this example with the help of an image.

There is a noise vector coming into the

forger who is generating fake wine.

Here the forger acts as a Generator.

The shop owner acts as a Discriminator.

The Discriminator gets two inputs; one is the fake wine, while the other is the real authentic wine. The shop owner has to figure out whether it is real or fake.

So, there are two primary components of Generative Adversarial Network (GAN) named:

Q118. You are given a dataset on cancer detection. You have built a classification model and achieved an accuracy of 96 percent. Why shouldn't you be happy with your model performance? What can you do about it?

Cancer detection results in imbalanced data. In an imbalanced dataset, accuracy should not be based as a measure of performance. It is important to focus on the remaining four percent, which represents the patients who were wrongly diagnosed. Early diagnosis is crucial when it comes to cancer detection, and can greatly improve a patient's prognosis.

Hence, to evaluate model performance, we should use Sensitivity (True Positive Rate), Specificity (True Negative Rate), F measure to determine the class wise performance of the classifier.

Q119. Which of the following machine learning algorithms can be used for inputting missing values of both categorical and continuous variables?

- K-means clustering
- Linear regression
- K-NN (k-nearest neighbor)
- Decision trees

The K nearest neighbor algorithm can be used because it can compute the nearest neighbor and if it doesn't have a value, it just computes the nearest neighbor based on all the other features.

When you're dealing with K-means clustering or linear regression, you need to do that in your pre-processing, otherwise, they'll crash. Decision trees also have the same problem, although there is some variance.

Q120. Below are the eight actual values of the target variable in the train file. What is the entropy of the target variable?

[0, 0, 0, 1, 1, 1, 1, 1]

Choose the correct answer.

Q121. What are the feature vectors?

A feature vector is an n-dimensional vector of numerical features that represent an object. In machine learning, feature vectors are used to represent numeric or symbolic characteristics (called features) of an object in a mathematical way that's easy to analyze.

Q122. What is root cause analysis?

Root cause analysis was initially developed to analyze industrial accidents but is now widely used in other areas. It is a problem-solving technique used for isolating the root causes of faults or problems. A factor is called a root cause if its deduction from the problem-fault-sequence averts the final undesirable event from recurring.

Q123. What is logistic regression?

Logistic regression is also known as the logit model. It is a technique used to forecast the binary outcome from a linear combination of predictor variables.

Q124. What is collaborative filtering?

Most recommender systems use this filtering process to find patterns and information by collaborating perspectives, numerous data sources, and several agents.

Q125. Do gradient descent methods always converge to similar points?

They do not, because in some cases, they reach a local minima or a local optima point. You would not reach the global optima point. This is governed by the data and the starting conditions.

Q126. What is the goal of A/B Testing?

This is statistical hypothesis testing for randomized experiments with two variables, A and B. The objective of A/B testing is to detect any changes to a web page to maximize or increase the outcome of a strategy.

Q127. What are the drawbacks of the linear model?

- The assumption of linearity of the errors
- It can't be used for count outcomes or binary outcomes
- There are overfitting problems that it can't solve

Q128. What is the law of large numbers?

It is a theorem that describes the result of performing the same experiment very frequently. This theorem forms the basis of frequency-style thinking. It states that the sample mean, sample variance and sample standard deviation converge to what they are trying to estimate.

Q129. What are the confounding variables?

These are extraneous variables in a statistical model that correlates directly or inversely with both the dependent and the independent variable. The estimate fails to account for the confounding factor.

Q130. What is star schema?

It is a traditional database schema with a central table. Satellite tables map IDs to physical names or descriptions and can be connected to the central fact table using the ID fields; these tables are known as lookup tables and are principally useful in real-time applications, as they save a lot of memory. Sometimes, star schemas involve several layers of summarization to recover information faster.

Q131. How regularly must an algorithm be updated?

You will want to update an algorithm when:

- You want the model to evolve as data streams through infrastructure
- The underlying data source is changing
- There is a case of non-stationarity

Q132. What are eigenvalue and eigenvector?

Eigenvalues are the directions along which a particular linear transformation acts by flipping, compressing, or stretching.

Eigenvectors are for understanding linear transformations. In data analysis, we usually calculate the eigenvectors for a correlation or covariance matrix.

Q133. Why is resampling done?

Resampling is done in any of these cases:

- Estimating the accuracy of sample statistics by using subsets of accessible data, or drawing randomly with replacement from a set of data points
- Substituting labels on data points when performing significance tests
- Validating models by using random subsets (bootstrapping, cross-validation)

Q134. What is survivorship bias?

Survivorship bias is the logical error of focusing on aspects that support surviving a process and casually overlooking those that did not because of their lack of prominence. This can lead to wrong conclusions in numerous ways.

Q135. How do you work towards a random forest?

The underlying principle of this technique is that several weak learners combine to provide a strong learner. The steps involved are:

Q136. What are the important skills to have in Python with regard to data analysis?

The following are some of the important skills to possess which will come handy when performing data analysis using Python.

- Good understanding of the built-in data types especially lists, dictionaries, tuples, and sets.
- Mastery of N-dimensional NumPy Arrays.
- Mastery of Pandas dataframes.
- Ability to perform element-wise vector and matrix operations on NumPy arrays.
- Knowing that you should use the Anaconda distribution and the conda package manager.
- Familiarity with Scikit-learn. **Scikit-Learn Cheat Sheet**
- Ability to write efficient list comprehensions instead of traditional for loops.
- Ability to write small, clean functions (important for any developer), preferably pure functions that don't alter objects.
- Knowing how to profile the performance of a Python script and how to optimize bottlenecks.

Credit: kdnuggets, Simplilearn, Edureka, Guru99, Hackernoon, Datacamp, Nitin Panwar, Michael Rundell

Deep Learning questions

Q1. Differentiate between AI, Machine Learning and Deep Learning.

Artificial Intelligence is a technique which enables machines to mimic human behavior.

Machine Learning is a subset of AI technique which uses statistical methods to enable machines to improve with experience.

Deep learning is a subset of ML which make the computation of multi-layer neural network feasible. It uses Neural networks to simulate human-like decision making.

Q2. Do you think Deep Learning is Better than Machine Learning? If so, why?

Though traditional ML algorithms solve a lot of our cases, they are not useful while working with high dimensional data, that is where we have a large number of inputs and outputs. For example, in

the case of handwriting recognition, we have a large amount of input where we will have a different type of inputs associated with different type of handwriting.

The second major challenge is to tell the computer what are the features it should look for that will play an important role in predicting the outcome as well as to achieve better accuracy while

doing so.

Q3. What is Perceptron? And How does it Work?

If we focus on the structure of a biological neuron, it has dendrites which are used to receive inputs. These inputs are summed in the cell body and using the Axon it is passed on to the next

biological neuron as shown below.

Dendrite:

Dendrite:

Receives signals from other neurons

Cell Body:

Cell Body:

Sums all the inputs

Axon:

Axon:

It is used to transmit signals to the other cells

Similarly, a perceptron receives multiple inputs, applies various transformations and functions and provides an output. A Perceptron is a linear model used for binary classi+cation. It models a

neuron which has a set of inputs, each of which is given a specific weight. The neuron computes some function on these weighted inputs and gives the output.

■

Subscribe

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Q4. What is the role of weights and bias?

What is the role of weights and bias?

For a perceptron, there can be one more input called

bias

bias

. While the weights determine the

slope

slope

of the classifier line, bias allows us to shift the line towards left or right. Normally bias is treated

as another weighted input with the input value x

Q5. What are the activation functions?

What are the activation functions?

Activation function translates the inputs into outputs. Activation function decides whether a neuron should be activated or not by calculating the weighted sum and further adding bias with it. The

purpose of the activation function is to introduce non-linearity into the output of a neuron.

There can be many Activation functions like:

Linear or Identity

Unit or Binary Step

Sigmoid or Logistic

Tanh

ReLU

Softmax

Q6. Explain Learning of a Perceptron.

Explain Learning of a Perceptron.

1

.

Initializing the weights and threshold.

2

.

Provide the input and calculate the output.

3

.

Update the weights.

4

.

Repeat Steps 2 and 3

Wj (t+1) – Updated Weight

Wj (t+1) – Updated Weight

Wj (t) – Old Weight

Wj (t) – Old Weight

d – Desired Output

d – Desired Output

y – Actual Output

y – Actual Output

x – Input

x – Input

Q7. What is the significance of a Cost/Loss function?

What is the significance of a Cost/Loss function?

A cost function is

a measure of the accuracy

a measure of the accuracy

of the neural network with respect to a given training sample and expected output. It provides the performance of a neural network as a whole. In

deep learning, the goal is to minimize the cost function. For that, we use the concept of gradient descent.

Q8. What is gradient descent?

What is gradient descent?

Gradient descent

is an optimization algorithm used to minimize some function by iteratively moving in the direction of steepest descent as defined by the negative of the gradient.

Stochastic Gradient Descent:

Stochastic Gradient Descent:

Uses only a single training example to calculate the gradient and update parameters.

Batch Gradient Descent:

Batch Gradient Descent:

Calculate the gradients for the whole dataset and perform just one update at each iteration.

Mini-batch Gradient Descent:

Mini-batch Gradient Descent:

Mini-batch gradient is a variation of stochastic gradient descent where instead of single training example, mini-batch of samples is used. It's one of the most

popular optimization algorithms.

Q9. What are the benefits of mini-batch gradient descent?

This is more efficient compared to stochastic gradient descent.

The generalization by finding the flat minima.

Mini-batches allows help to approximate the gradient of the entire training set which helps us to avoid local minima.

0.

Q10.What are the steps for using a gradient descent algorithm?

Q10.What are the steps for using a gradient descent algorithm?

Initialize random weight and bias.

Pass an input through the network and get values from the output layer.

Calculate the error between the actual value and the predicted value.

Go to each neuron which contributes to the error and then change its respective values to reduce the error.

Reiterate until you find the best weights of the network.

Q10. What are the shortcomings of a single layer perceptron?

Well, there are two major problems:

Single-Layer Perceptrons cannot classify non-linearly separable data points.

Complex problems, that involve a lot of parameters cannot be solved by Single-Layer Perceptrons

Q11. What is a Multi-Layer-Perceptron

A multilayer perceptron (MLP) is a deep, arti+cial neural network. It is composed of more than one perceptron. They are composed of an input layer to receive the signal, an output layer that makes

a decision or prediction about the input, and in between those two, an arbitrary number of hidden layers that are the true computational engine of the MLP.

Q12. What are the different parts of a multi-layer perceptron?

Input Nodes

Input Nodes

: The Input nodes provide information from the outside world to the network and are together referred to as the "Input Layer". No computation is performed in any of the Input

nodes – they just pass on the information to the hidden nodes.

Hidden Nodes:

Hidden Nodes:

The Hidden nodes perform computations and transfer information from the input nodes to the output nodes. A collection of hidden nodes forms a "Hidden Layer". While a network

will only have a single input layer and a single output layer, it can have zero or multiple Hidden Layers.

Output Nodes:

Output Nodes:

The Output nodes are collectively referred to as the "Output Layer" and are responsible for computations and transferring information from the network to the outside world.

Q13. What Is Data Normalization And Why Do We Need It?

Data normalization is very important preprocessing step, used to rescale values to +t in a speci+c range to assure better convergence during backpropagation. In general, it boils down to

subtracting the mean of each data point and dividing by its standard deviation.

These were some basic Deep Learning Interview Questions. Now, let's move on to some advanced ones.

Advance Interview Questions

Q14. Which is Better Deep Networks or Shallow ones? and Why?

Both the Networks, be it shallow or Deep are capable of approximating any function. But what matters is how precise that network is in terms of getting the results. A shallow network works with

only a few features, as it can't extract more. But a deep network goes deep by computing efficiently and working on more features/parameters.

Q15. Why is Weight Initialization important in Neural Networks?

Weight initialization is one of the very important steps. A bad weight initialization can prevent a network from learning but good weight initialization helps in giving a quicker convergence and a

better overall error.

```
params = [weights_hidden, weights_output, bias_hidden, bias_output]
def sgd(cost, params, lr=0.05):
grads = T.grad(cost=cost, wrt=params)
updates = []
for p, g in zip(params, grads):
updates.append([p, p - g * lr])
return updates
updates = sgd(cost, params)
```

Biases can be generally initialized to zero. The rule for setting the weights is to be close to zero without being too small.

Q16. What's the difference between a feed-forward and a backpropagation neural network?

A Feed-Forward Neural Network is a type of Neural Network architecture where the connections are "fed forward", i.e. do not form cycles. The term "Feed-Forward" is also used when you input

something at the input layer and it travels from input to hidden and from hidden to the output layer.

Backpropagation is a training algorithm consisting of 2 steps:

Feed-Forward the values.

Calculate the error and propagate it back to the earlier layers.

So to be precise, forward-propagation is part of the backpropagation algorithm but comes before back-propagating.

Q17. What are the Hperparameteres? Name a few used in any Neural Network.

Hyperparameters are the variables which determine the network structure(Eg: Number of Hidden Units) and the variables which determine how the network is trained(Eg: Learning Rate).

Hyperparameters are set before training.

Number of Hidden Layers

Network Weight Initialization

Activation Function

Learning Rate

Momentum

Number of Epochs

Batch Size

Q18. Explain the different Hyperparameters related to Network and Training.

Network Hyperparameters

Network Hyperparameters

The number of Hidden Layers:

The number of Hidden Layers:

Many hidden units within a layer with regularization techniques can increase accuracy. Smaller number of units may cause underfitting.

Network Weight Initialization:

Network Weight Initialization:

Ideally, it may be better to use different weight initialization schemes according to the activation function used on each layer. Mostly uniform distribution is used.

Activation function:

Activation function:

Activation functions are used to introduce nonlinearity to models, which allows deep learning models to learn nonlinear prediction boundaries.

Training Hyperparameters

Training Hyperparameters

Learning Rate:

Learning Rate:

The learning rate de+nes how quickly a network updates its parameters. Low learning rate slows down the learning process but converges smoothly. Larger learning rate speeds up

the learning but may not converge.

Momentum:

Momentum:

Momentum helps to know the direction of the next step with the knowledge of the previous steps. It helps to prevent oscillations. A typical choice of momentum is between 0.5 to 0.9.

The number of epochs:

The number of epochs:

Number of epochs is the number of times the whole training data is shown to the network while training. Increase the number of epochs until the validation accuracy

starts decreasing even when training accuracy is increasing(overfitting).

Batch size:

Batch size:

Mini batch size is the number of sub-samples given to the network after which parameter update happens. A good default for batch size might be 32. Also try 32, 64, 128, 256, and so

on.

Q19. What is Dropout?

Dropout is a regularization technique to avoid over+tting thus increasing the generalizing power. Generally, we should use a small dropout value of 20%-50% of neurons with 20% providing a good

starting point. A probability too low has minimal effect and a value too high results in under-learning by the network.

Use a larger network. You are likely to get better performance when dropout is used on a larger network, giving the model more of an opportunity to learn independent representations.

Q20. In training a neural network, you notice that the loss does not decrease in the few starting epochs. What could be the reason?

The reasons for this could be:

The learning is rate is low

Regularization parameter is high

Stuck at local minima

Q21. Name a few deep learning frameworks

TensorFlow

Caffe

The Microsoft Cognitive Toolkit/CNTK

Torch/PyTorch

MXNet

Chainer

Keras

Q22. What are Tensors?

Tensors are nothing but a de facto for representing the data in deep learning. They are just multidimensional arrays, that allows you to represent data having higher dimensions. In general, Deep

Learning you deal with high dimensional data sets where dimensions refer to different features present in the data set.

Q23. List a few advantages of TensorFlow?

It has platform flexibility

It is easily trainable on CPU as well as GPU for distributed computing.

TensorFlow has auto differentiation capabilities

It has advanced support for threads, asynchronous computation, and queue es.

It is a customizable and open source.

Q24. What is Computational Graph?

A computational graph is a series of TensorFlow operations arranged as nodes in the graph. Each node takes zero or more tensors as input and produces a tensor as output.

Basically, one can think of a Computational Graph as an alternative way of conceptualizing mathematical calculations that takes place in a TensorFlow program. The operations assigned to different

nodes of a Computational Graph can be performed in parallel, thus, providing better performance in terms of computations.

Q25. What is a CNN?

Convolutional neural network (CNN, or ConvNet) is a class of deep neural networks, most commonly applied to analyzing visual imagery. Unlike neural networks, where the input is a vector, here

the input is a multi-channeled image. CNNs use a variation of multilayer perceptrons designed to require minimal preprocessing.

Q26. Explain the different Layers of CNN.

There are four layered concepts we should understand in Convolutional Neural Networks:

Convolution:

Convolution:

The convolution layer comprises of a set of independent +lters. All these +lters are initialized randomly and become our parameters which will be learned by the network subsequently.

ReLu:

ReLu:

This layer is used with the convolutional layer.

Pooling:

Pooling:

Its function is to progressively reduce the spatial size of the representation to reduce the number of parameters and computation in the network. Pooling layer operates on each feature

map independently.

Full Connectedness:

Full Connectedness:

Neurons in a fully connected layer have full connections to all activations in the previous layer, as seen in regular Neural Networks. Their activations can hence be computed

with a matrix multiplication followed by a bias offset.

Q27. What is an RNN?

Recurrent Networks are a type of arti+cial neural network designed to recognize patterns in sequences of data, such as text, genomes, handwriting, the spoken word, numerical times series data.

Recurrent Neural Networks use

backpropagation

backpropagation

algorithm for training Because of their

internal memory

internal memory

, RNN's are able to remember important things about the input they received, which

enables them to be very precise in predicting what's coming next.

Q28. What are some issues faced while training an RNN?

Recurrent Neural Networks use backpropagation algorithm for training, but it is applied for every timestamp. It is commonly known as

Back-propagation Through Time

Back-propagation Through Time

(BTT).

There are some issues with Back-propagation such as:

Vanishing Gradient

Exploding Gradient

Q29. What is Vanishing Gradient? And how is this harmful?

When we do Back-propagation, the gradients tend to get smaller and smaller as we keep on moving backward in the Network. This means that the neurons in the Earlier layers learn very slowly as

compared to the neurons in the later layers in the Hierarchy.

Earlier layers in the Network are important because they are responsible to learn and detecting the simple patterns and are actually the building blocks of our Network.

Obviously, if they give improper and inaccurate results, then how can we expect the next layers and the complete Network to perform nicely and produce accurate results. The Training process

takes too long and the Prediction Accuracy of the Model will decrease.

Q30. What is Exploding Gradient Descent?

Exploding gradients are a problem when large error gradients accumulate and result in very large updates to neural network model weights during training.

Gradient Descent process works best when these updates are small and controlled. When the magnitudes of the gradients accumulate, an unstable network is likely to occur, which can cause poor

prediction of results or even a model that reports nothing useful what so ever.

Q31. Explain the importance of LSTM.

Long short-term memory(LSTM) is an arti+cial recurrent neural network architecture used in the +eld of deep learning. Unlike standard feedforward neural networks, LSTM has feedback

connections that make it a "general purpose computer". It can not only process single data points, but also entire sequences of data.

They are a special kind of Recurrent Neural Networks which are capable of learning long-term dependencies.

Q32. What are capsules in Capsule Neural Network?

Capsules

are a vector specifying the features of the object and its likelihood. These features can be any of the instantiation parameters like position, size, orientation, deformation, velocity, hue,

texture and much more.

A capsule can also specify its attributes like angle and size so that it can represent the same generic information. Now, just like a neural network has layers of neurons, a capsule network can have

layers of capsules.

Now, let's continue this Deep Learning Interview Questions and move to the section of autoencoders and RBMs.

Q33. Explain Autoencoders and it's uses.

An

autoencoder

neural network is an Unsupervised Machine learning algorithm that applies backpropagation, setting the target values to be equal to the inputs. Autoencoders are used to reduce

the size of our inputs into a smaller representation. If anyone needs the original data, they can reconstruct it from the compressed data.

Q34. In terms of Dimensionality Reduction, How does Autoencoder differ from PCAs?

An autoencoder can learn non-linear transformations with a non-linear activation function and multiple layers.

It doesn't have to learn dense layers. It can use convolutional layers to learn which is better for video, image and series data.

It is more efficient to learn several layers with an autoencoder rather than learn one huge transformation with PCA.

An autoencoder provides a representation of each layer as the output.

It can make use of pre-trained layers from another model to apply transfer learning to enhance the encoder/decoder.

Q35. Give some real-life examples where autoencoders can be applied.

Image Coloring:

Image Coloring:

Autoencoders are used for converting any black and white picture into a colored image. Depending on what is in the picture, it is possible to tell what the color should be.

Feature variation:

Feature variation:

It extracts only the required features of an image and generates the output by removing any noise or unnecessary interruption.

Dimensionality Reduction:

Dimensionality Reduction:

The reconstructed image is the same as our input but with reduced dimensions. It helps in providing a similar image with a reduced pixel value.

Denoising Image:

Denoising Image:

The input seen by the autoencoder is not the raw input but a stochastically corrupted version. A denoising autoencoder is thus trained to reconstruct the original input from the

noisy version.

Q36. what are the different layers of Autoencoders?

An Autoencoder consist of three layers:

Encoder

Code

Decoder

Q37. Explain the architecture of an Autoencoder.

Encoder:

Encoder:

This part of the network compresses the input into a latent space representation. The encoder layer encodes the input image as a compressed representation in a reduced dimension.

The compressed image is the distorted version of the original image.

Code:

Code:

This part of the network represents the compressed input which is fed to the decoder.

Decoder:

Decoder:

This layer decodes the encoded image back to the original dimension. The decoded image is a lossy reconstruction of the original image and it is reconstructed from the latent space

representation.

Q38. What is a Bottleneck in autoencoder and why is it used?

The layer between the encoder and decoder, ie. the code is also known as Bottleneck. This is a well-designed approach to decide which aspects of observed data are relevant information and what

aspects can be discarded.

It does this by balancing two criteria:

Compactness of representation, measured as the compressibility.

It retains some behaviourally relevant variables from the input.

Q39. Is there any variation of Autoencoders?

Convolution Autoencoders

Sparse Autoencoders

Deep Autoencoders

Contractive Autoencoders

Q40. What are Deep Autoencoders?

The extension of the simple Autoencoder is the Deep Autoencoder. The +rst layer of the Deep Autoencoder is used for +rst-order features in the raw input. The second layer is used for second-

order features corresponding to patterns in the appearance of first-order features. Deeper layers of the Deep Autoencoder tend to learn even higher-order features.

A deep autoencoder is composed of two, symmetrical deep-belief networks:

First four or five shallow layers representing the encoding half of the net.

The second set of four or five layers that make up the decoding half.

Q41. What is a Restricted Boltzmann Machine?

Restricted Boltzmann Machine

is an undirected graphical model that plays a major role in Deep Learning Framework in recent times.

It is an algorithm which is useful for dimensionality reduction, classification, regression, collaborative filtering, feature learning, and topic modeling.

Q42. How Does RBM differ from Autoencoders?

Autoencoder is a simple 3-layer neural network where output units are directly connected back to input units. Typically, the number of hidden units is much less than the number of visible ones.

The task of training is to minimize an error or reconstruction, i.e. find the most efficient compact representation for input data.

RBM shares a similar idea, but it uses stochastic units with particular distribution instead of deterministic distribution. The task of training is to +nd out how these two sets of variables are actually